

Heterogeneous Purchases, Heterogeneous Multipliers

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Abstract

Do the effects of local fiscal multipliers vary based on the type of government purchases? We find that government purchases of services generate larger increases in local employment, labor income, and output than purchases of goods. This variation stems from key factors: differences in sectoral characteristics—particularly markups—between good-producing and service-producing industries; distinct impacts on business turnover and productivity; and greater tradeability of goods, which amplifies spillover effects beyond the local area. Our results highlight important mechanisms shaping the effectiveness of fiscal policy across spending categories.

JEL Codes: E10, E62, H57, R12

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1 Introduction

Until recently, the literature on fiscal multipliers treated government spending as a homogeneous good—an abstract aggregate collapsing tanks, hospital equipment, and cloud-computing contracts into the same unit of analysis.¹ This simplification, inherited from national accounting conventions, prevents us from assessing how the composition of government purchases shapes the transmission of fiscal shocks. While recent research has explored heterogeneity in private-sector responses, it has largely overlooked differences within government spending itself. This omission is increasingly important in an economy where both final consumers and producers have increased their demands for services at the expense of goods.²

Our paper fills this gap by examining whether the composition of government purchases influences the size and nature of fiscal multipliers—an issue long neglected due to the scarcity of granular spending data (Chodorow-Reich, 2019). Specifically, we answer two questions. First, does the size of the local fiscal multipliers depend on the type of government purchases? Second, what mechanisms explain any heterogeneity? To answer these, we compile a dataset covering 40 years of U.S. military procurement at the contract level. The dataset identifies the exact product or service purchased by the government, enabling us to classify purchases into goods and services. We exploit variation in procurement patterns across metropolitan statistical areas (MSAs) and over time to quantify the causal effects of the two types of public procurement purchases on employment, labor income and output (Nakamura and Steinsson, 2014).

Budget allocation remains a policy-driven decision, whereas firm characteristics primarily affect the transmission of demand shocks. By isolating the multiplier effects associated with the allocation of purchases between the two categories of spending, we can more cleanly identify variation in the multiplier that is attributable to policymakers’ spending composition rather than to sector-specific (or firm-specific) characteristics. A better understanding of fiscal policy implications associated with the two categories of spending can assist policymakers in optimizing public resource allocation.

We first document a shift in the composition of government defense spending during the early 1990s. Before this period, over 65% of defense procurement purchases were directed towards goods. However, in recent decades, the US federal government has allocated similar shares of resources to both goods and services. This pattern highlights the relevance of our paper given this secular trend in the composition of government purchases.

Second, we estimate employment, labor income and GDP local fiscal multipliers by category of purchases. An increase of one-percentage-point in services purchases (normalized by personal income) results in an on-impact employment growth of 0.25%. This impact strengthens over time, reaching 0.87% at the three-year horizon. Contrarily, employment responses for good purchases are positive but not statistically significant across any time horizon following the spending. These multipliers range from 0.03% on-impact to 0.12% at

¹Cox et al. (2024) and Bouakez et al. (2025) are two recent works that made the case of considering spending as heterogeneous.

²In 2020, the service sector comprised almost 80% of the US production. (Herrendorf et al., 2014) document that services dominate household consumption, and (Francois and Hoekman, 2010; Duarte and Restuccia, 2010) show that services became the main intermediate inputs used by firms in production.

the four-year horizon, constituting a quarter to a tenth of the estimates for services spending. Notably, unlike good purchases, service purchases stimulate significant increases in household labor income, with increases between 0.31% and 1.53% at different horizons. Importantly, within a constrained sample due to limitations in data availability, an increase of one-percentage-point in service purchases yields a GDP response that varies between 0.3 on-impact and reaches the peak of 1.8 at the three-year horizons. In contrast, the GDP responses associated with the purchase of goods are not statistically significant, exhibiting values that range from approximately no effect on-impact to around 0.1 at the two-year horizon.

To gauge a better sense of the size of the employment estimates, we analyze the newly-created jobs for \$100,000 spent in procurement contracts. The number of new jobs created substantially differs between the two types of purchases. \$100,000 in procurement contracts that acquire services create 0.58 new jobs at impact, rising to 1.71 after two years, while good purchases only yield 0.08 new jobs initially and 0.15 after two years. These findings align with [Chodorow-Reich \(2019\)](#), who reports job creation estimates between 0.76 and 3.93 for the two-year horizon.

We also investigate whether employment effects concentrate solely within industries directly benefiting from procurement purchases or if the effects spill to non-receiving sectors. We observe that both purchase categories generate positive direct impacts on the leading recipients of such expenditure.³ Moreover, our findings indicate that indirect effects, manifesting as positive employment responses in sectors different from the leading recipient, are primarily associated with service purchases. The surge in manufacturing sector employment after service purchases is substantial, whereas other sectors witness moderate effects. Good purchases, however, do not trigger significant employment responses outside the manufacturing sector. This observation aligns with the growing use of services as intermediate inputs ([Duarte and Restuccia, 2010](#)).

We turn next to the causes of different responses for services and goods purchases. We explore four avenues: differences in producers’ characteristics, extensive margin and productivity gains, tradeability, and locality-specific characteristics. We find that producers’ characteristics, tradeability and the extensive margin play an important role in determining the heterogeneous multiplier effects, while locality-specific characteristics do not seem to contribute to these differences.

We explore the role of producers’ characteristics —contribution to final consumption, contribution to investment, centrality in the production network, labor intensity, markup, and price rigidity —identified by previous studies as important drivers of the fiscal multiplier ([Bouakez et al., 2023](#); [Bouakez et al., 2025](#); [Cox et al., 2024](#)). First, we document that a larger share of service purchases compared to good purchases is allocated to producers in industries with a larger contribution to final consumption and investment, higher centrality —upstream sectors that supply intermediate inputs to other industries —and higher labor intensity. Conversely, a greater share of good purchases compared to service purchases are channeled towards contractors operating in sectors with higher markups, and higher frequency of price adjustments. Motivated by these differences, we assess the influence of these industry-attributes on the different fiscal responses after

³Manufacturing firms secure approximately 80% of good-purchase contracts, while contracts in service industries receive 55% of service purchases.

the two categories of purchases. Our empirical findings suggest that markups play a central role in explaining part of the local multiplier responses to goods and services purchases.

Government purchases may also generate demand-driven growth by facilitating firm dynamism and productivity gains. We document that service purchases boost the establishment’s entry and exit into sectors with high labor productivity growth. By contrast, goods purchases increase the entry and exit rates in sectors with low labor productivity growth, consistent with the fact that goods purchases are directed to contractors in higher markup sectors. These results suggest that service purchases generate an aggregate improvement in labor productivity larger than goods purchases because firm turnover mostly occurs in high-growth sectors. These results provide suggestive evidence that the extensive margin and productivity gains might amplify differently the fiscal responses to the two categories of purchases.

As goods are more tradeable than services, one would expect that the production of goods spills into neighboring locations, and small local responses after good purchases may be due to geographic spillovers. For this reason, we quantify the “outflow” effects on neighboring MSAs.⁴ Nevertheless, a significant portion of trade activity extends beyond neighboring MSAs. Given that the geographic sub-contracting becomes a minor issue at the state level, we compare the estimates from the regressions using MSA-level data with those using state-level data. The empirical findings provide mixed evidence about the role of geographic spillovers, and we cannot completely rule out the possibility that geographic spillovers contribute to explain part of the gap between the estimates for the two categories of purchases.

By contrast to the previous channels, we do not find support for locality-specific characteristics to matter in explaining the differences in the responses between categories of purchases. The allocation of spending in the two categories is uneven across locations, with some receiving a relatively larger share of spending in one of the two categories than the other. The differential geographical dispersion of goods-producing and services-producing firms may present a challenge for isolating the differential effects of goods-versus-services spending from differential effects arising from economic differences in the geographies in which spending occurs. We address this concern by calculating the correlations between the share of spending and location-specific characteristics that matter in quantifying the size of the fiscal response, such as the share of small firms (Juarros, 2020) or the MSA-size. We find these correlations are close to zero, implying no relationship between the share of a type of purchase and location-specific characteristics.

Our empirical analysis provides suggestive evidence that multiple channels, operating simultaneously, contribute in explaining the differences between the estimates for services and goods purchases.⁵ The categorization between services and goods captures the joint effect of all these channels. Furthermore, from a policy standpoint, the more relevant and observable distinction for governments lies in the decision to allocate spending toward goods versus services, rather than between sectors (or firms) with different intrinsic characteristics, not directly observable by the policymaker. Intrinsic characteristics are not only more difficult to

⁴We define neighboring MSAs as those located within a 50 miles radius.

⁵For a precise quantification of the relative importance of each of these channels, one would need a structural model, which is beyond the scope of this study.

measure in real time, but also less actionable in terms of fiscal design. In contrast, distinguishing between goods and services is both conceptually straightforward and operationally meaningful for public procurement decisions, and policymakers. For these reasons, we believe that the goods-versus-services distinction provides clearer and more policy-relevant insights that help the design of fiscal interventions by emphasizing a dimension of spending that is both empirically tractable and directly tied to government decision-making.

We contribute to the literature by documenting the differences in fiscal multipliers by the type of spending, whereas previous research has emphasized heterogeneity in economic conditions. [Alonso \(2017\)](#) and [Dupor et al. \(2021\)](#) study the effect of fiscal policy on the composition of private consumption. Our paper investigates the other side of the coin. Instead of exploring the heterogeneity in the consumption responses to shocks, we focus on the heterogeneity in the shocks themselves, i.e., service- and good- purchases. We show this heterogeneity is an important yet overlooked determinant of the spending multiplier. [Boehm \(2020\)](#) is a notable exception and studies the different output responses to government investment and consumption shocks.⁶ We classify spending into good and service purchases rather than consumption and investment. The importance of this classification is twofold. First, the US economy has shifted from a good economy to a service economy. Studying the differences between these two types of spending would help policymakers in designing effective fiscal interventions. Second, goods and services use different production technologies. Quantifying their effects separately helps to identify important determinants of the response to fiscal stimuli. Our findings provide complementary insights to those already documented in the literature.⁷

Our paper also relates to the vast literature on the determinants of the fiscal multiplier. This literature has highlighted the role of the business cycle ([Riera-Crichton et al., 2015](#); [Suárez-Serrato and Wingender, 2016](#); [Buchheim et al., 2020](#)), trade openness ([Ilzetzi et al., 2013](#); [Corbi et al., 2019](#)), the exchange rate regime ([Born et al., 2013](#)), population demographics ([Basso and Rachedi, 2021](#)), labor market rigidity ([Cole and Ohanian, 2004](#); [Hall, 2009](#); [Gorodnichenko et al., 2012](#)), automatic stabilizers ([Dolls et al., 2012](#); [Galeano et al., 2021](#)), public indebtedness ([Ilzetzi et al., 2013](#)), the degree of monetary policy accommodation ([Woodford, 2011](#)), firm-size distribution ([Juarros, 2020](#)), and the direction of the intervention ([Barnichon et al., 2022](#)) in amplifying the response of economic activity to public spending. This paper concentrates on the characteristics of the fiscal spending itself rather than the characteristics of the economy.⁸ Our findings show that the composition of the basket of products purchased leads to different multiplier effects.

Lastly, our paper is linked to the growing literature that studies, in the presence of heterogeneous agents, the aggregate effects of fiscal policy and its channels. This literature has investigated both the heterogeneity

⁶For developing countries, [Ilzetzi et al. \(2013\)](#) show that distinguishing between government consumption (goods and services) and investment is an important determinant of the impact of government spending shocks on output.

⁷Another complementary branch of the literature is the literature on infrastructure multiplier. The infrastructure multiplier literature is not suitable for addressing the short economic impact of fiscal policies because infrastructure projects typically have a long lead time. Furthermore, infrastructure spending tends to focus on a single industry (construction). Unlike the infrastructure multiplier literature, the purchase multiplier literature explores the effect of fiscal policies directed at already established products. These differences allow us to study the short-term effects of fiscal policy and quantify the role of industry characteristics in determining differences in the responses to fiscal policy.

⁸Studies on the infrastructure multiplier also focus on the characteristics of the spending, but they do not simultaneously test the heterogeneity in the responses coming from different types of spending.

on the consumer side (Galí et al., 2007; Brinca et al., 2016; Hagedorn et al., 2019), the heterogeneity on the production side (Bouakez et al., 2023; Bouakez et al., 2025), and the heterogeneity on both sides (Flynn et al., 2021). Our results show that the aggregate fiscal multiplier is higher when they originate in the service sectors, as they generate stronger general-equilibrium (indirect) effects. We test the importance of several industries’ characteristics in quantifying the employment response after fiscal spending. In this regard, our findings are complementary to Bouakez et al. (2025) that use a structural approach to establish the importance of sectors’ characteristics in determining the aggregate effect of sector-specific fiscal interventions.

2 Data

2.1 Government Procurement Contracts

We assemble a dataset on military procurement contracts awarded by the US Department of Defense (DoD). We harmonize military procurement contracts data from two sources: the National Archives and Records Administration (NARA) for the period 1966-2006, and USASpending.gov for the period 2007-2019.⁹ The data from both sources are based on DD-350 and DD-1057 military procurement forms that account for approximately 96% of contracts awarded by the DoD.¹⁰

The data contain detailed information, including the contract identification number, the dates of action and completion, the transaction value, the location where the contract is performed, and the Product Service Codes (PSCs) or the Federal Supply Codes (FSCs). We construct the government procurement spending in the following way. We define the year in which the government purchase occurs as the year of the signature date, which occurs when the contract is either awarded or modified.¹¹ The modifications of existing contracts could consist of downward revisions of the contract obligation. These modifications are reported as negative entries. As in Auerbach et al. (2020), we consider contracts with obligations and de-obligations with magnitudes within 0.5% of each other to be null and void. For comparability over time, we convert the nominal transaction value into real values using the US Bureau of Labor Statistics’ Consumer Price Index.¹²

The Product Service Codes and the Federal Supply Codes play a crucial role in classifying government purchases as either service or good purchases. These codes consist of a 4-digit alphanumeric classification

⁹The data from USASpending.gov are available from 2001. The period 2001-2006 validates the quality of the NARA’s data.

¹⁰Appendix A.1 tests the similarity of our data to that used in previous studies. Our universe of military spending is very similar to one from previous studies in both the aggregate level and the geographic distribution.

¹¹The government fiscal year has been defined from October 1st to September 30th since 1976. The mismatch between the fiscal year of the government and the calendar year could cause a time inconsistency between military spending and other economic variables. Thus, we use the calendar year as a reference year.

¹²No official statistics accurately measure local differences in prices. The BEA has recently started to produce MSA-level GDP deflators. These deflators are constructed by applying national price indices to current dollar values of MSA-level GDP at the industry level. Hence, these statistics do not capture local differences in prices but in industry composition. Some recent works, such as Stroebel and Vavra (2019) or Hazell et al. (2022) have constructed better measures for local price indexes. Unfortunately, we cannot use these measures either because they cover different levels of geographic aggregation and time periods or because they exploit proprietary microdata, which we don’t have access to. The lack of reliable official local price indexes and the difficulty of constructing better indexes have led other researchers, for example, Auerbach et al. (2020) or Bisgaard-Larsen et al. (2025), to use nominal prices. In our case, as we exploit a much longer time series than previous studies, the use of nominal prices may induce serious measurement errors. Thus, we opted for deflating the nominal values by national price indexes. Furthermore, goods and services may face different price adjustments. As a robustness analysis, we estimate our main specification by using two different aggregate deflators for goods and services spending.

representing the type of deliverable requested by the contract. Although some codes have been added or deleted over time, the existing codes refer to the same deliverable starting from 1979. The Product Service Codes refer to a service as the deliverable, and the first digit is a letter. The Federal Supply Codes request goods as deliverables, and the first digit is a number. Thus, if the first digit is a letter, we classify that purchase as “Purchase of Services.” If the first digit is a number, we classify that spending as “Purchase of Goods.” To our knowledge, previous studies have not used this data feature in their analysis.¹³

Finally, the identification strategy exploits the geographic variation in government purchases. As standard in the literature, the geographic allocation is based on the location where the tasks of the contracts are performed. The detailed location information allows us to geolocate contracts in narrow geographic areas. The available information differs between our two sources of data. In the USASpending.gov data, we know the city and the zip code of the performing firm.¹⁴ We use these two pieces of information to identify the county where a firm operates. In NARA, the county where a firm performs the contract is reported. We use the spatial crosswalks provided by the National Bureau of Economic Research (NBER) to aggregate the county-level contracts into Metropolitan Statistical Areas (MSAs).

2.2 Economic Outcomes

We use three outcome variables to proxy changes in economic activity after a local purchase shock. The first dependent variable is employment, measured in the number of jobs.¹⁵ The employment figures are collected from the Bureau of Economic Analysis (BEA).¹⁶ The second outcome variable consists of salary and wages, and proxies for the labor income. These data are collected annually from the BEA.¹⁷ The third variable is the Gross Domestic Product (GDP). Different from the previous outcome variables, GDP at the MSA level is only available starting from 2001. Finally, we collect personal income from the BEA. Personal income measures households’ income from salary and wages, Social Security and other government programs, dividends and interest, business ownership, and other sources to normalize government spending.

2.3 Sample Construction

We construct the final sample by applying two sets of filters. From the universe of DoD procurement contracts, we drop those with missing information in at least one of these dimensions: the locality in which the contract is performed, the year in which the contract is signed, the product code,¹⁸ or the industry in

¹³Appendix A.2 reports the major product codes included in the two categories.

¹⁴Following Demyanyk et al. (2019), if we know that a contract has been performed in the US, but we do not know the exact location where it was performed, we assign the location of the contract recipient as the performance location. Notice that we adjust a marginal share of contracts differently from Demyanyk et al. (2019). That is the case because we locate contracts not only using the postal code but also the city. There are contracts for which information about the postal code is missing, but the information about the city is not.

¹⁵The employment figures cover the total of full-time and part-time workers in any industry, including government and government enterprises.

¹⁶The BEA estimates of employment and wages differ from the BLS data because BEA adjusts for employment and wages not covered, or not fully covered, by the state unemployment insurance (UI) and the unemployment compensation for Federal employees (UCFE) programs. First, BEA adds estimates of employment and wages to the BLS data to bridge small gaps in UI coverage. Second, BEA uses additional source data to estimate employment and wages. Third, BEA adjusts employment and wages for misreporting under the UI and UCFE programs.

¹⁷For consistency, salary and wages, as all other nominal variables, are converted into real terms to absorb price variations.

¹⁸Our analysis starts from 1979 because the product codes before 1979 are not consistent with those after that date.

which the contractor operates. We apply these filters because the absence of any of these variables prevents us from correctly assigning contracts to one of the spending categories. We restrict the sample to 2019 to avoid the effect of the COVID-19 pandemic on our estimates.

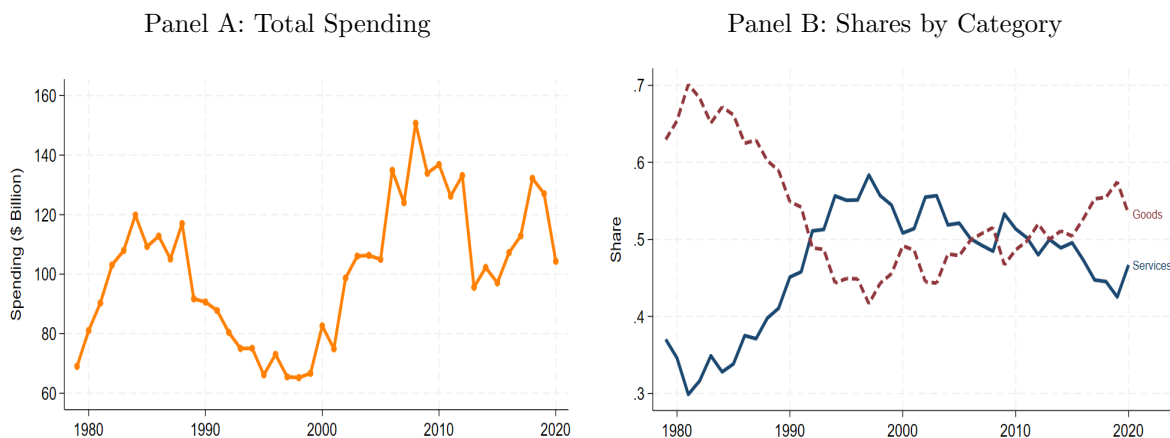
The second set of filters is applied to the data after aggregating purchases by MSA. These filters are used to minimize the effect of outliers or misreported values on our estimates. We exclude MSAs with incomplete histories in any outcome variable or with growth rates between two consecutive periods greater than 100% or smaller than -50% . We remove MSAs with an average population smaller than 50,000 inhabitants. We only include MSAs with non-negative ratios of aggregate purchases to personal income smaller than 1.5.¹⁹ Finally, as in Auerbach et al. (2020) and Demyanyk et al. (2019), the analysis is carried out at the locality aggregated level rather than per capita.

The final sample of about 28 million of procurement contracts allocated to 334 MSAs from 1979 to 2019. The data filters have a minor impact on the representativeness of the procurement contracts.

3 Descriptive Statistics

Figure 1 reports spending over 1979 – 2019. Panel A shows the evolution of total spending. There are three sharp rises in spending. The first increase is in the 1980s as a consequence of the Reagan military buildup, the second is in the early 2000s due to the Afghan and Iraqi wars, and the last is in recent years, driven by the escalations in military buildups and competition with Russia and China.

Figure 1: Procurement Spending



Notes: The national level statistics are calculated by aggregating the microdata on military procurement contracts available from NARA and USASpending. The face value in the DoD procurement contracts is deflated as explained in Section 2.1. The classification of the purchases into goods and services is based on the Federal Product Classification. The figures are computed using the 344 MSAs in the final sample.

Panel B reports the shares of purchases by category. The division of the federal procurement purchases across the two categories highlights a reallocation of federal government purchases from goods to services

¹⁹We initially started the construction of the sample considering Core-based statistical areas (CBSAs) rather than MSAs. The filters, particularly the minimum location size in terms of population, leave in the sample only a handful of micropolitan statistical areas. Therefore, we decided to restrict the analysis only to MSAs.

between the 1980s and the 1990s. As the product classification has been harmonized over the sample period, changes in the shares are not due to changes in the product definition. Until the beginning of the 1980s, the spending on goods was about 60% of the procurement contracts. Since the 1990s, the spending has been split between goods and services with similar shares.²⁰ Nevertheless, as showed in Table 1, the DoD awarded most of its procurement contracts (91%) to purchase goods, and only a small share (9%) is directed to the acquisition of services, implying that service-purchase contracts are larger than good-purchase contracts.²¹

An interesting comparison is between the composition of government spending and private consumption expenditures. Figure B.1 shows that the share of private expenditures in services has also increased. Private consumers started to reallocate their spending from goods to service purchases at the beginning of the 1970s. Furthermore, the comparison highlights that government spending is more goods-driven than private consumption. This finding adds to the results of Cox et al. (2024) that emphasize the existence of a sectoral bias in government spending, by showing that the importance of goods and services purchases differs substantially between government spending and private consumption.

Table 1: Descriptive Statistics

	All	Goods	Services
Share of Spending (%)	-	54	46
Share of Contracts (%)	-	91	9
Average Value (Thousand USD)	173	102	860
90%-to-50%-percentile Ratio	39	34	21
Share of Spending for Top 1% (%)	88	92	66

Note: The national level statistics are calculated by aggregating the microdata on military procurement contracts available from NARA and USASpending.gov. DoD procurement contracts are deflated, as explained in Section 2.1. The classification of the spending into goods and services is based on the Federal Product Classification. The statistics are computed using the 344 MSAs in the final sample.

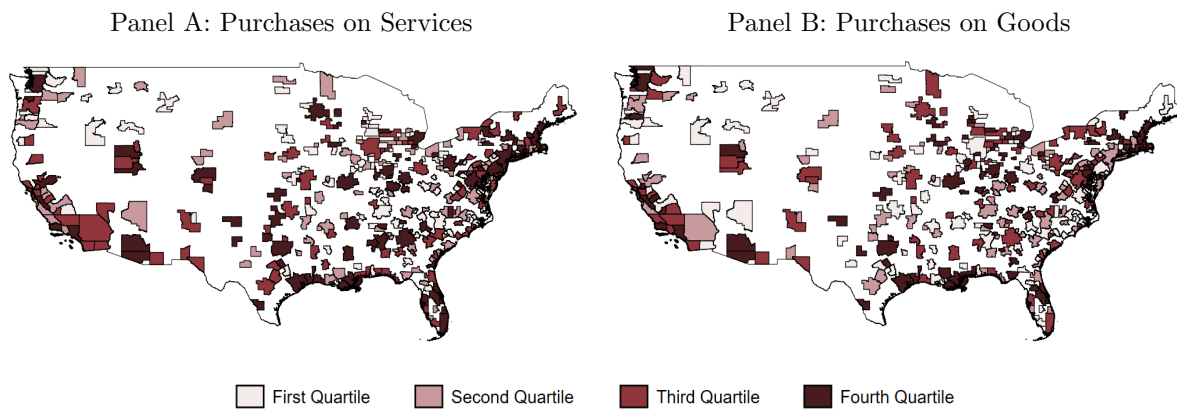
The last two rows of Table 1 explore the distributional characteristics of the purchases within categories. The distribution of contracts for the purchase of goods has a significantly fatter right-tail than the distribution for spending on services. In the case of spending for goods, contracts at the top decile of the distribution are

²⁰A substantial portion of DoD spending is tied to traditional defense and military procurement. The DoD has also played a critical role in supporting technologies and services that, shortly after their creation, diffused broadly across the civilian economy. Well-known examples include the development of the internet, GPS, and medical innovations targeting public health threats such as endemic diarrheal diseases and viral infections. These cases highlight how DoD investments have historically anticipated and contributed to broader structural changes in the economy. To provide additional context, Table A.2 presents the list of product codes classified by spending category. These codes span a wide range of economic activities beyond core defense functions. To quantify changes over time, we calculate the growth in spending shares by category between two decades: 1981–1990 and 1991–2000. Among goods-related categories, 44 out of 78 experienced a decline in their share of total DoD spending. In contrast, 20 out of 24 service-related categories saw an increase in their share. The largest increases in services were in categories Q - Medical Services, F - Natural Resources Management, and B - Special Studies/Analysis. These categories were negligible in the early 1980s but grew significantly over time.

²¹Services purchases includes R&D activities. R&D contracts are usually large. Removing R&D contracts from the service spending, the differences between the two categories are less marked. Nevertheless, the differences persist because the distribution of the two types of contracts differs, with service purchases more concentrated in the top tail of the distribution.

about 40 times greater than the median contract. The 90%-to-50%-percentile ratio is smaller for spending on services, with the contracts at the top decile is 21 times greater than the median contract. Among contracts to produce goods, the top 1% of contracts accounts for 92% of the goods spending. This result implies that although the majority of contracts buy goods, the largest bulk of the spending is allocated to a relatively small subset. The share of services spending allocated to the top percentile is 66% of the services spending, implying a more equal allocation of spending across contracts. Our findings are complementary to Cox et al. (2024), which show that spending is granular and concentrated among a few firms. Their analysis is based on the contract recipients. The granularity in recipients does not necessarily imply that the actual performance of the tasks is granular. Our results highlight that, in addition to granularity in contract recipients, there exists a granularity in contract performance.²² Similarly to Cox et al. (2024), we also document substantial variation in the range of contract values, and we emphasize significant differences in the distributional features of contract size across categories of spending.

Figure 2: Geographic Distribution of the Procurement Spending



Notes: The quartile to which an MSA belongs is assigned based on the average military spending in real terms that the MSA receives over 1979 – 2019. The classification of the spending into goods and services is based on the Federal Product Classification. The figures are computed using the 344 MSAs in the final sample.

As the identification strategy exploits the cross-sectional variation in purchases, Figure 2 explores the geographic heterogeneity in spending allocation across MSAs. The figure shows the quartile to which an MSA belongs based on the average value of the military spending it received over 1979 – 2019. There are two main results. First, military spending is unequally distributed across MSAs. The top 40 MSAs in terms of awarded spending account for more than 35% and 45% of spending on goods and services, respectively. Most MSAs that receive the largest share of DoD spending are located along the two coastal regions and the Midwest. Second, the geographic allocation of spending differs between the two categories. Only 35% of MSAs are in the same quartile for both categories. The heat-maps suggest that most MSAs in the Midwest are in the top two quartiles of spending on goods, and only a few of these MSAs are ranked as high in the

²²We cannot directly test this granularity in contract performers because the data do not report the name of the firm that performs a contract. We only observe the place of performance.

distribution of services spending. This result mirrors the geographic economic structure of the US.²³

These descriptive statistics provide evidence that although the DoD signs a large number of contracts, the largest share of government procurement spending is captured by a handful of contracts. Furthermore, these contracts are not equally distributed across MSAs. These findings imply that the distribution of spending at both contract and MSA levels is skewed to the right.

4 Empirical Strategy

4.1 Specification

Our empirical strategy builds on the works of Nakamura and Steinsson (2014), Dupor and Guerrero (2017), and Auerbach et al. (2020). We exploit cross-sectional variation in military procurement purchase shares and time-series variation in national DOD spending to estimate impulse response functions of economic activity when local government purchases change.²⁴ The benchmark specification is defined as follows:

$$\frac{v_{l,t+k} - v_{l,t-1}}{v_{l,t-1}} = \beta^k \frac{G_{l,t+k} - G_{l,t-1}}{Y_{l,t-1}} + \alpha_l^k + \delta_{t+k} + \varepsilon_{l,t+k} \quad (1)$$

where v is an outcome of interest in MSA l at horizon $t + k$ with $k = \{0, \dots, 4\}$. The endogenous variable $G_{l,t+k} - G_{l,t-1}$ measures the change in military spending normalized by the first lag of the personal income in location l , $Y_{l,t-1}$. The specification also includes MSA-fixed effects α_l^k to control for MSA-specific trends, and the time-fixed effects δ_{t+k} to account for any mechanical correlation between secular trends in military spending and unobserved local factors.²⁵ The main coefficient of interest is β^k , which quantifies the change in the outcome of interest (employment, labor income, or GDP growth rate) in a window of k years when government purchase increases by a one percentage point of MSA's personal income.²⁶

We augment the benchmark specification to examine the heterogeneous effects of the different government spending components. In this respect, we estimate the specification (2):

$$\frac{v_{l,t+k} - v_{l,t-1}}{v_{l,t-1}} = \beta_g^k \frac{G_{l,t+k}^g - G_{l,t-1}^g}{Y_{l,t-1}} + \beta_s^k \frac{G_{l,t+k}^s - G_{l,t-1}^s}{Y_{l,t-1}} + \alpha_l^k + \delta_{t+k} + \varepsilon_{l,t+k} \quad (2)$$

where G^g represents the purchase of goods, and G^s the purchase of services, with $G^g + G^s = G$ for each period and locality.

²³The Midwest has the highest concentration of production occupations with an average employment share in production jobs, almost 50% higher than the US average.

²⁴In our estimation, the temporal dimension plays a more important role, particularly for the computation of the shift shares. As we disaggregated the spending by sub-category, the within-locality volatility of the disaggregated spending is much higher than that of the total spending. Thus, we need longer periods to compute representative local-level shares.

²⁵Standard errors are clustered by MSA.

²⁶Our empirical strategy provides estimates for the open economy relative local fiscal multiplier. Estimates of the open economy relative local fiscal multiplier cannot be straightforwardly translated into a national multiplier. This is the case for two reasons. First, spillover effects could propagate outside the locality's borders. Second, different assumptions and parameterizations of macroeconomic models could generate a broad range of estimates of the national multiplier.

4.2 Instrumental variables research design

Procurement spending is potentially endogenous due to political or economic factors. The observed changes in procurement spending may respond to unobserved local shocks, also affecting the outcome variables. For example, firms in locations with more contracts may have exerted more effort through lobbying to win those contracts. Our estimate may be biased if such firm-specific shocks affect the outcome variables.

We follow the instrumental variable research design proposed by [Nakamura and Steinsson \(2014\)](#) to obtain causal estimates of the fiscal multipliers. This approach consists of constructing a shift-share instrument in the spirit of [Bartik \(1991\)](#). The instrument is obtained by combining the nationwide changes in procurement spending with a measure of comparative advantage that certain MSAs have in obtaining contracts. The instrument, $Z_{l,t+h}$, is defined as:

$$Z_{l,t+h} = \kappa_l \frac{G_{t+k} - G_{t-1}}{Y_{l,t-1}} \quad (3)$$

where κ_l is the average share of spending in MSA l . As the empirical specifications are in changes and we use MSA-fixed effects, the conditional variation provided by the instrument comes from national-level changes in military spending, which is plausibly exogenous to local shocks that may affect the outcome variables.

The instruments for equation (2) are defined as:

$$Z_{l,t+h}^g = \kappa_l^g \frac{G_{t+k}^g - G_{t-1}^g}{Y_{l,t-1}}; \quad Z_{l,t+h}^s = \kappa_l^s \frac{G_{t+k}^s - G_{t-1}^s}{Y_{l,t-1}} \quad (4)$$

with Z^g being the instrument for changes in goods spending and Z^s the instrument for changes in services spending. κ_l^g and κ_l^s measure the shares of spending in goods and services captured by MSA l , respectively.

4.3 Identification assumptions and threats

Although this instrumental variables research design has become the gold standard in the literature that studies cross-sectional fiscal multipliers, it is not free of identification threats that may bias our estimates. Our identification assumption is conditional on the MSA and time-fixed effects. The instrument is not systematically associated with any unobserved political or economic characteristics that may explain the outcomes of interest. This identification strategy may suffer from three main identification threats.

The first identification threat is related to measurement errors induced by outsourcing military procurement. A contract is assigned to its place of performance, defined as where the product is assembled or processed. Suppose sub-contractors outside the place of performance are used in the intermediate steps of production. In that case, we would geographically misallocate part of the outsourced military spending. This measurement error will create an attenuation bias in our estimates of the local fiscal multiplier. For the purpose of our analysis, a more serious threat in interpreting the relative size of the product-spending multipliers would arise if the measurement error is systematically correlated with a specific spending category. Section 6.3 discusses this concern.

The second threat to identification is the potential presence of geographical spillovers. This would imply

a violation of the stable unit treatment value assumption. More concretely, this violation would arise if defense spending in a specific MSA affects neighboring MSAs without contracts. The importance of the bias in our estimates depends on the direction of the geographic spillovers. The spillovers are positive when the demand for final consumption or intermediate goods increases in the neighboring MSAs due to input-output linkages. In this case, our estimates would suffer from a downward bias. When the increase in spending affects factor prices and the allocation of production factors across MSAs, the spillovers could be negative, and our estimates would suffer from an upward bias. Section 6.3 discusses this concern.

The third concern relates to the drawbacks of shift-share instruments (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022; Adao et al., 2019). Our empirical strategy uses a shift-share instrumental variable with several shifts (e.g., as many as $G_{t+k} - G_{t-1}$ differences can be computed in the data) and one single share for each MSA (e.g., the time-invariant measure of comparative advantage). Borusyak et al. (2022) show the shift-share instrumental variable requires either the shares or the shifts to be uncorrelated with unobserved characteristics that may affect the local outcomes. In our view, the shifts are a reasonable source of quasi-random variation because military buildups respond to international geopolitical events rather than to unobserved factors such as automation, trade competition, or national fiscal policy, which could have a heterogeneous impact across MSAs. The exogeneity of the shifts assures the validity of the instrument, even when MSAs with a high comparative advantage in attracting procurement contracts experience a different economic trend than MSAs with a low comparative advantage. As both outcomes and instruments are measured in changes, our research design purges the instrument from the potential correlation between the shares and the unobserved local economic trends. Furthermore, the locality-fixed effects absorb any MSA-specific secular trend. Thus, the variation provided by the instrument consists of deviations in spending from its long-term trend, which is plausibly exogenous to local economic characteristics.

5 Results

In this section, we first investigate the local outcome responses to local total government procurement spending. Then, we present the responses in the local outcome variables to government procurement spending divided by the two categories of interest —services and goods. In the last part of the section, we implement a set of checks to assess the robustness of our empirical findings.

5.1 Main Results

We start by estimating the response of employment, labor income, and output to local total government procurement spending.²⁷ The estimates in Panel A of Table B.1 suggest a positive and significant effect on employment. Indeed, we estimate an effect of 0.091 on-impact and 0.268 at the two-year horizon. Similarly to Panel A, Panel B also shows a positive and significant effect of procurement spending on labor income at any horizon. A one-percentage-point increase in government spending increases labor income by 0.13%

²⁷These estimates have been called in different ways in the literature. For example, Nakamura and Steinsson (2014) refers to it as the “open economy relative multiplier.” By contrast, Auerbach et al. (2020) refer to it as the “local fiscal multiplier.” In the remainder of the paper, we will use the notation introduced by Auerbach et al. (2020).

on-impact and above 0.51% after a two-year horizon. Finally, Panel C reports the output response to shocks to the total spending. We document a positive but non-significant effect of government spending on GDP on-impact, but a positive and significant effect of 0.4% at the two-year horizon.²⁸

Now, we turn to the core results of the paper. We explore the effects of the two types of government spending: spending on goods and services. Figure 3 plots the estimated effects reported in Table B.2. Panel A shows the effect on employment after goods and services spending shocks. We observe a remarkable difference between the two patterns. On the one hand, the estimates after a shock in services spending are positive and statistically significant. A one-percentage-point increase in normalized services spending by personal income generates an increase in employment of 0.24% on-impact and 0.72% two periods after the shock. On the other hand, goods spending shocks have small effects on the employment multiplier. The estimates are statistically non-significant and range between 0.03% at impact and 0.07% at the two-year horizon, respectively. We test whether the gaps between two sets of coefficients are statistically different from each other, and we find that we reject the null hypothesis that the two estimates are equal.

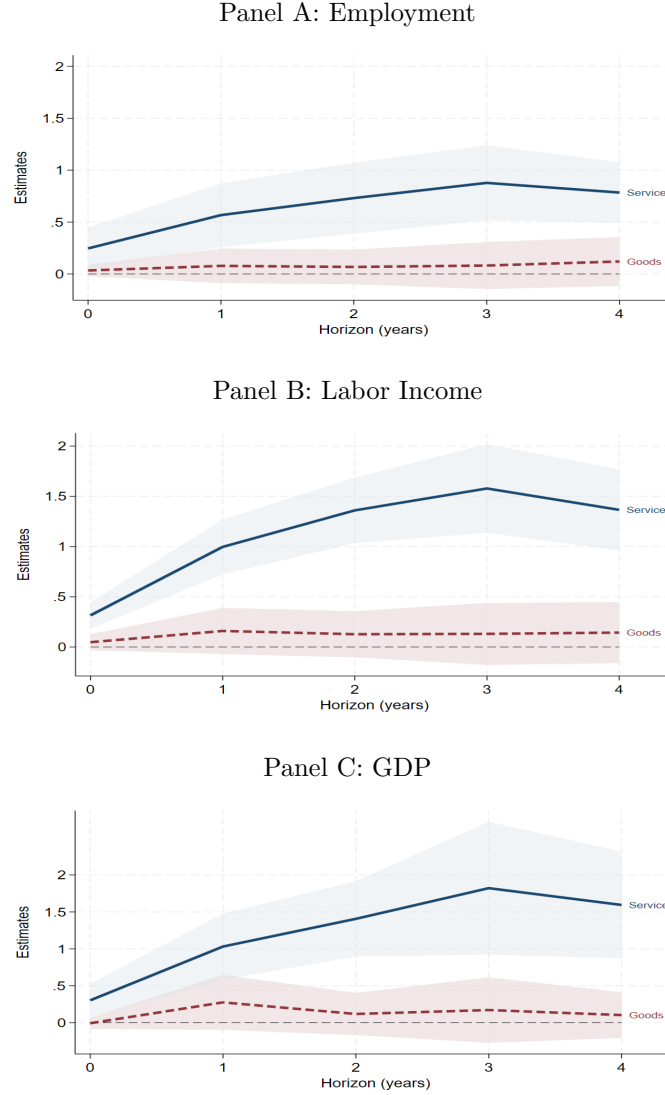
The increase in the demand for products leads firms to hire more workers. As firms hire more workers (or similarly, workers work more hours), the labor income increases, and households can increase their consumption. The increased demand for products further increases labor demand. [Alonso \(2017\)](#) and [Bouakez et al. \(2025\)](#) show that the direct effect on the output of government spending shocks, consisting of output creation due to changes in government demand, is relatively small. Most of the aggregate impact comes from the indirect effect, consisting of changes in households' demand due to increases in their income as a consequence of government spending shocks. Our results suggest that this virtuous circle only activates after shocks to services spending but not goods spending. In Panel B of Figure 3 we show that there is a sizable increase in wages and salaries after a shock to services spending. Specifically, a one-percentage-point increase in normalized services spending generates a rise in labor income up to 1.3% within two periods after the shock. By contrast, labor income only slightly increases after goods spending shocks.²⁹ Finally, the last panel of Figure 3 shows the GDP impulse responses. Similarly to the previous two panels, we document a statistically positive effect of services spending on GDP, while the effect on GDP after goods spending shocks is close to zero and non-statistically significant at any standard levels.

If firm ownership goes largely beyond MSA boundaries, cross-MSA capital income may make the dividend income “less local.” As good producers rely on capital more than service producers, the spill of dividends

²⁸We compare our estimates to [Auerbach et al. \(2020\)](#), which is the closest paper to ours in terms of specification and sample construction. Our employment and income responses at the two-year horizon are 0.258 and 0.505, respectively. [Auerbach et al. \(2020\)](#) report 0.189 and 0.335 for employment and income, respectively. While the estimates for employment and income are relatively close, the discrepancy for the estimates in GDP is much larger. Indeed, we report 0.4 at $t + 2$, and [Auerbach et al. \(2020\)](#) 1.1 at the same horizon. The discrepancy is mostly due to the different geographic aggregation. [Auerbach et al. \(2020\)](#) focuses their analysis on CBSAs, while we restrict the sample to MSAs. Extending our analysis to CBSAs, we get an estimate for GDP at $t + 2$ of 0.9, which is close to the coefficient of [Auerbach et al. \(2020\)](#). Furthermore, if we consider states instead of MSAs, we get an estimate of 1.2 at $t + 2$, which is close to 1.4 documented by [Nakamura and Steinsson \(2014\)](#). Overall, we conclude that the differences in estimates can be explained by the different geographic aggregation.

²⁹Although not reported, the responses to personal income closely track the responses to labor income, with the only difference being that the effect size is smaller for personal income. This stems from a combination of changes in labor income and dividends. We check the responses of dividends to the two types of shocks, and we find that firms do not significantly adjust dividends paid to households after spending shocks.

Figure 3: Impulse Responses - Purchase by Category



Notes: The panel includes 344 MSAs and consists of 12,024 observations for the period 1979 – 2019 for Panels A and B, and 4,676 observations for the period 2001 – 2019 for Panel C. Standard errors are clustered by MSA. The shaded areas represent the 90% confidence intervals.

to other locations may be more relevant for spending on goods and may partially explain the limited effects that we document after goods spending shocks. Although tracking ownership across MSA borders is hard, [Kolko and Neumark \(2010\)](#) show that 92.3% of establishments are locally-owned. Furthermore, [Mygrant \(2020\)](#) considers a subset of US firms and shows that the shares held within 20 miles of a firm establishment are 100% for family-owned businesses and 80% for employee-owned businesses. Only for publicly-traded firms that account for less than 1% of all US firms, the shares held within a 20 mile radius from a firm establishment are less than a quarter. These figures about the ownership structure of US businesses suggest that the spill of dividend income into neighboring locations is a relatively small issue. We further address

this concern by quantifying the effect of both spending shocks on dividends in the neighboring locations.³⁰ We find no effect on dividends in neighboring locations after either of the two types of spending, reinforcing the previous conclusion that the spill of dividend income into neighboring locations is relatively marginal.

As highlighted in [Auerbach et al. \(2020\)](#), DoD spending consists of two components: unanticipated spending associated with a new production —“demand shock” —and anticipated contract obligations that do not cause new production —“wealth transfers.” The negligible effect of DoD procurement spending on goods could be due to a weak first-stage instrument for goods spending. The weak instrument would not allow us to isolate “demand shocks” from “wealth transfers” effectively. Alternatively, large goods producers may smooth production over fluctuations in national DoD spending more than services producers. Both cases would lead to underestimating the fiscal effects of goods spending. In Table B.2, we report the conditional first-stage F-statistics for the validity of the instruments. Since the core of the paper is testing whether the multipliers differ across types of spending, we need to show that we can identify both coefficients separately. This requires demonstrating that both instruments are valid separately for identifying their respective coefficients. We test whether a particular endogenous regressor alone is identified by implementing the test developed by [Sanderson and Windmeijer \(2016\)](#) that is based on the “partialled-out” procedure proposed by [Angrist and Pischke \(2009\)](#).³¹ The 10% critical value for the conditional first-stage F-statistics for the validity of the instruments in a perfectly-identified model with two endogenous variables is 7.03. The reported F-statistics emphasize that we can reject the hypothesis that the instruments are weak, minimizing the concern of not effectively identifying “demand shocks.”

The specifications in equations (1) and (2) ignore the dynamics of spending and the outcome variables for the years between the periods of study. In other words, when the change in spending is measured as the difference in spending between $t + k$ and $t - 1$, the two specifications ignore the changes in the years $t \in \{t - 1, t + k\}$. We follow [Ramey and Zubairy \(2018\)](#) and estimate a specification where the outcome and spending variables are defined as the cumulative change between $t - 1$ and $t + k$. We refer to the cumulative response of local outcome variables to the cumulative local spending as the “cumulative local fiscal multiplier.” We construct the dependent variables as the cumulative change in an outcome variable between $t - 1$ and $t + k$:

$$\sum_{h=0}^k \frac{v_{l,t+h} - v_{l,t-1}}{v_{l,t-1}},$$

and similarly, the cumulative local spending as the sum of the spending by category that a location receives between $t - 1$ and $t + k$:

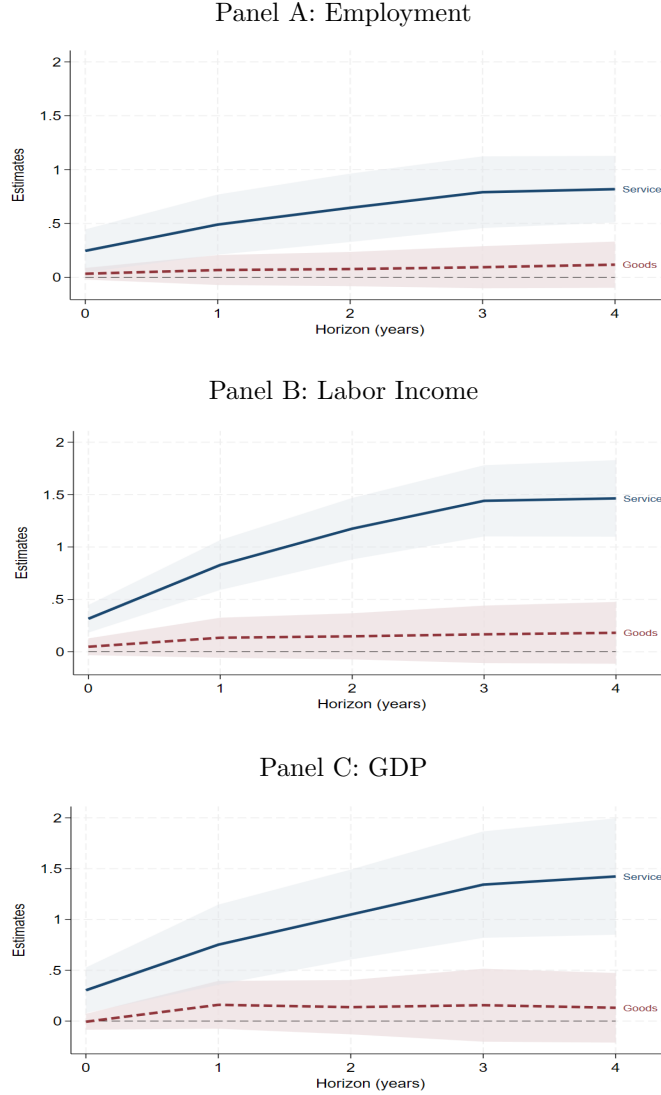
$$\sum_{h=0}^k \frac{G_{l,t+h}^i - G_{l,t-1}^i}{Y_{l,t-1}} \quad \forall i \in \{s, g\}.$$

The estimates showed in Figure 4 come from Table B.3. The estimates highlight a stark difference between goods and services spending. Specifically, we document positive and significant responses of the

³⁰The results are not reported but are available upon request.

³¹We also implement tests using only goods spending as an endogenous regressor. In all these cases, the F-statistics is above the Stock-Yogo critical value of 10, implying we can reject the presence of weak instruments.

Figure 4: Cumulative Fiscal Multiplier



Notes: The balanced panel includes 344 MSAs and consists of 12,024 observations for the period 1979 – 2019 for Panels A and B, and 4,676 observations for the period 2001 – 2019 for Panel C. Standard errors are clustered by MSA. The shaded areas represent the 90% confidence intervals.

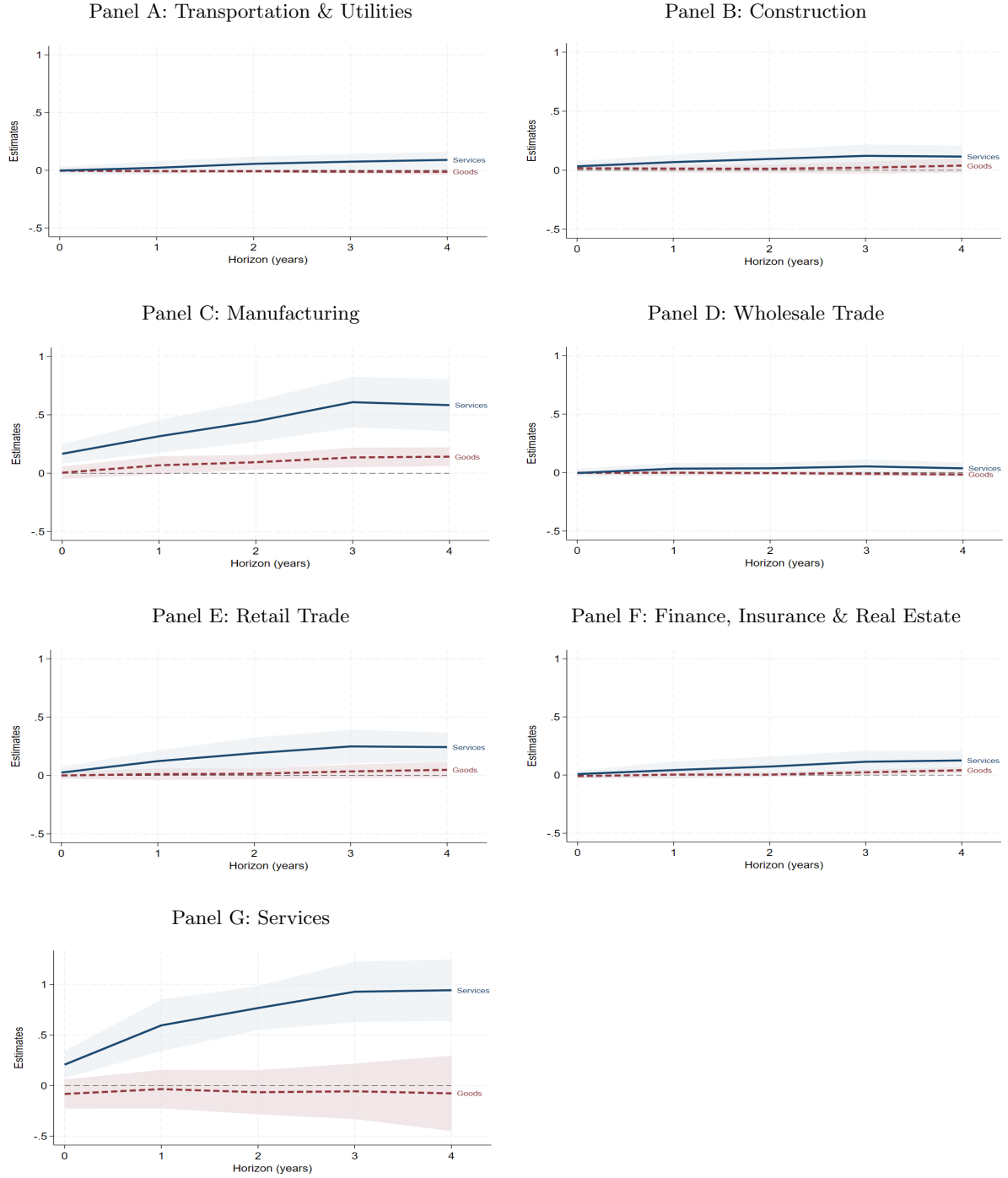
cumulative local outcome variables to the cumulative local spending in services. By contrast, the cumulative effects in response to the cumulative local spending in goods are positive but non-significant. Furthermore, the increases in GDP and labor income are greater than one after the two-year horizon.

We also investigate whether the employment responses occur in the industries that receive the procurement spending or the effects spill over to industries that do not receive the spending. To explore this aspect, we collect employment data by county and industry,³² and we aggregate them by MSAs and major NAICS groupings.³³ As industries differ in their size, for better comparability across industries, we report the

³²Eckert et al. (2020) develop an algorithm to impute several suppressed employment by industry from the County Business Patterns. The suppression is due to confidentiality risks. Data are available until 2016.

³³We use the grouping reported by Nakamura and Steinsson (2014). We exclude Agriculture and Mining because several

Figure 5: Newly-created Jobs by Industry



Notes: All panels report the estimates from equation (2) for different industries separately. The instruments are computed as in equation (4). The balanced panel consists of 12,024 observations and includes 344 MSAs for the period 1979 – 2019. Standard errors are clustered by MSA. The shaded areas represent the 90% confidence intervals.

newly-created jobs rather than employment responses. The number of newly-created jobs per \$100,000 in MSAs have missing employment figures for these sectors due to confidentiality risks.

procurement spending is calculated as

$$\Delta E_{t+k} = \hat{\beta}_i^k \frac{1}{TN} \sum_{t=1980}^T \sum_{l=1}^N \frac{100,000}{Y_{l,t-1}} E_{l,t-1} \quad \forall i \in \{s, g\}.$$

with $\hat{\beta}_i^k$ representing the estimate at different horizons k of the specification (2) estimated for each industry. $E_{l,t-1}$ is the level of employment in MSA l the year before the spending shock, $Y_{l,t-1}$ is the one year lag of the MSA's personal income, and T and N are the number of years and MSAs in our sample, respectively.

The estimates for the newly-created jobs are showed in Table B.4. Before describing the industry results, we briefly discuss the estimates for the entire economy.³⁴ The number of new jobs created substantially differs between the two types of spending. \$100,000 in procurement contracts that acquire services create 0.58 new jobs at impact, and 1.71 after two years. By contrast, \$100,000 used to purchase goods generate 0.07 new jobs at impact and 0.15 after two years. Figure 5 summarizes the calculations for the two types of spending by industry. First, our estimates suggest both types of spending generate positive and significant direct effects on the industry that receives most of that spending.³⁵ Specifically, goods spending shocks generate 0.09 new jobs at the two-year horizon in the manufacturing sector, and services spending shocks create 0.77 new jobs after two years in the services sector. Furthermore, for both types of spending, the employment responses are the largest in the industry that receives the largest share of the procurement contracts. Second, our findings suggest that indirect effects of positive employment responses that spill to other sectors occur mainly after services spending. Indeed, the employment response after services spending is large in the manufacturing sector and moderate in the other industries. By contrast, goods spending shocks do not generate significant employment responses in sectors other than manufacturing. The presence of indirect effects, which materialize as positive employment responses spilling to other sectors only after services spending shocks, is consistent with the fact that the services share of intermediate inputs has risen over time (Duarte and Restuccia, 2010).

5.2 Robustness

In the benchmark results in Table B.2, we deflate both types of spending by the same aggregate price index. Nevertheless, it is plausible that goods and services face different price adjustments across locations. To fully address this concern, we would need MSA-specific price indexes for goods and services that are unavailable. Thus, we provide some evidence that different price adjustments for goods and services do not matter in determining our main result by deflating the goods and services procurement spending series by the aggregate goods and services deflators.³⁶ Table B.5 highlights the same pattern as in the benchmark

³⁴Briganti et al. (2025) show that the estimates of cost-per-job per year based on the employment multipliers derived from DoD procurement contracts are larger than those associated with other types of government spending.

³⁵About 80% of the goods spending is allocated to contractors belonging to the manufacturing sector, and 55% of the services spending is directed to contractors in the services industry.

³⁶We collect price indexes for services, durable and non-durable goods from the U.S. Bureau of Labor Statistics. We construct the price index for goods by weighted averaging durable and non-durable goods price indexes with weights equal to the relative weights of each consumption basket.

specification. Specifically, we document a positive and significant effect after service spending in all three outcome variables and a non-significant and close to zero effect after goods spending.

Another concern relates to the size distribution of government procurement contracts. Government procurement spending is granular, and a relatively small share of contracts captures a large share of the total spending. As contracts are concentrated in a small number of firms, our results could be driven by the nature of the specific contractors rather than the differences between goods and services. Furthermore, as we showed in Section 3, services contracts are, on average, larger than goods contracts. Therefore, the size of the individual contracts may cause the differential fiscal multipliers rather the category of spending. We address these concerns by taking three different approaches that help ensure that our estimates are not disproportionately driven by large federal procurement contracts.

The first approach consists of excluding from the regression analysis all contracts in the top 1% of the value distribution within each MSA-year. We adopt this conservative approach for two main reasons. First, the top 1% filter ensures that we remove genuinely large contracts, regardless of whether they are for goods or services, avoiding the potential bias of targeting only specific categories. Second, defining “large” relative to the MSA-year level accounts for substantial variation in contract size across both geography and time. A contract of average national size might be unusually large for a small MSA in a given year, and vice versa. Similarly, average contract size can vary significantly over time. Since our empirical strategy relies on both cross-sectional and temporal variation, we use a local and time-specific threshold in applying this restriction. The estimates excluding contracts in the top 1% of the value distribution within each MSA-year reported in Table B.6 reinforce the previous conclusion.

We acknowledge that the exclusion of contracts from our analysis may potentially introduce a bias in our estimates. This exclusion reduces the spending at the MSA-year level while leaving the outcome variables unchanged, potentially leading to an overestimation of the effect of each dollar spent through procurement contracts. To address this concern, we implement a complementary strategy that aims to mitigate this bias. Specifically, we exclude MSAs that receive “large” procurement contracts over the period 1979–2019. We define a “large” contract as one exceeding one million USD in real terms, and we exclude MSAs where such contracts represent more than 1% of total number of procurement contracts during the sample period.³⁷ The estimates excluding the MSAs with a large share of large contracts are showed in Table B.7. We find that removing MSAs with a large share of large contracts does not drive the differences between the service and good spending multipliers. If anything, we document that the gap between the estimates for the two categories of spending is greater when we restrict the sample of MSAs.

Finally, we adopt a third strategy to tackle the concern with the large contracts. We explicitly control for the lagged average contract sizes at the MSA level by type of spending.³⁸ This specification allows us to assess the extent to which differences in responses between goods and services purchases are driven by the

³⁷We have also conducted robustness checks using alternative thresholds for both the contract size used to define a “large” procurement and the percentage used to classify MSAs with negligible exposure.

³⁸We normalize the lagged average contract size by the lagged personal income at the MSA level

larger size of service contracts. The coefficients controlling for the average contract size by type of spending are described in Table B.8. We document positive effects on the outcome variables due to service spending and no effects due to good spending. These results rule out the hypothesis that the differences in fiscal multiplier estimates between goods and services purchases are driven by differences in contract size.

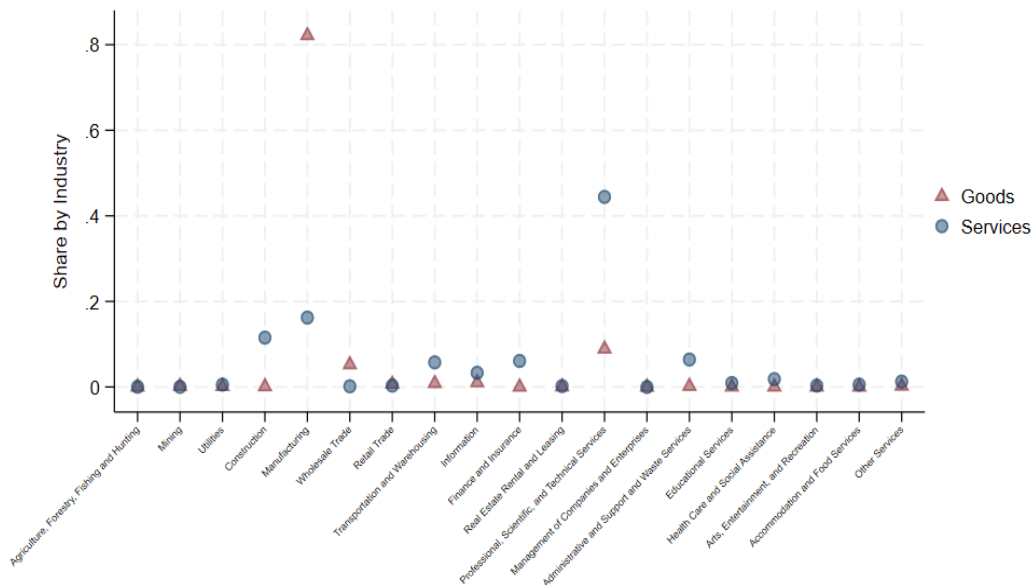
6 Mechanisms

Spending on goods and services differs along several dimensions. This section investigates several mechanisms that could explain the heterogeneity in the previous estimates by category of spending.

6.1 Industry-specific characteristics

Firms differ in characteristics and use different production methods, which can influence the impact of spending differently. Especially if good-producing and service-producing firms vary in these characteristics, it could lead to diverse amplifications of the two types of spending. Our data lacks information about the firm characteristics and production inputs. As a workaround, we construct industry-level metrics for these characteristics —contribution to final consumption, contribution to investment, centrality in the production network, labor intensity, markup, and price rigidity. We then allocate these metrics to contracts based on the respective industry of the contractor.

Figure 6: Share of Spending by Sectors



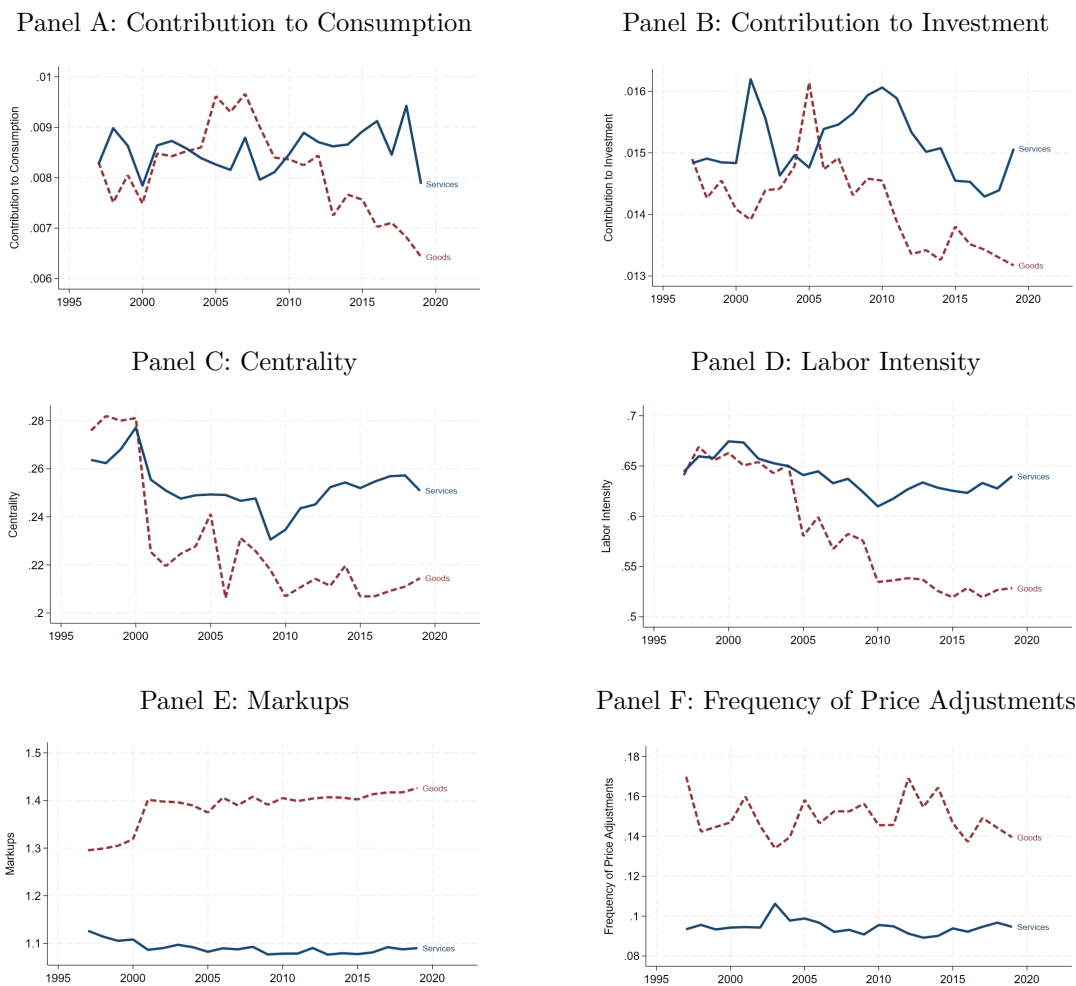
Notes: The classification of the spending into goods and services is based on the Federal Product Classification. The figures are computed using the 334 MSAs in the final sample.

Figure 6 shows the distribution of the two categories of spending by 2-digit NAICs codes. Goods spending is predominantly awarded to contractors that operate in the “*Manufacturing*” sector (80%), while services spending spans various sectors, with the “*Professional, Scientific, and Technical Services*” receiving about

40% of the DoD spending on services.³⁹ The marked differences in the sectoral allocation of the two types of spending make it important to examine whether the industries receiving these procurement contracts also differ in their characteristics.

Each panel of Figure 7 presents the weighted average by the two types of spending of one of the six sectoral characteristics —contribution to consumption, contribution to investment, centrality in the production network, labor intensity, markup, and price rigidity. The weights are constructed as the share of spending going to a specific industry relative to the total spending by category. These statistics highlight that firms receiving the two types of spending operate in industries that significantly differ along these characteristics.

Figure 7: Industry Confounders by Spending Category



Notes: The classification of the spending into goods and services is based on the Federal product classification. The figures are computed using the spending allocated to the 334 MSAs in the final sample.

Panel A shows the contribution to final consumption. Sectors that substantially contribute to the final demand for consumption are typically larger in size, and are better equipped to accommodate the increase in public spending without crowding-out private demand. We measure the contribution to final consump-

³⁹Auerbach et al. (2020) shows a similar ordering of industries receiving contracts without separating by products codes.

tion demand of a sector as the share of private consumption expenditure served by that specific industry. Our empirical findings show that firms receiving services spending operate in industries that, before 2010, contribute less to private consumption demand than firms receiving goods spending. In the recent years, while the contribution of sectors in which awardees of service contracts operate has remained constant, the contribution of sectors in which good-producing contracts operate has declined rapidly.

Panel B presents the contribution to investment. Investment goods often require longer and more complex production processes, delaying the output response. Furthermore, investment-producing firms are often capital-intensive and may operate near full capacity, requiring additional investment in capital stock to fulfill the contract requirements. If firms face credit constraints or financing frictions, they will not be able to quickly adjust their capital stock. Overall, these theoretical insights point out to a lower output response in industries with a relatively large contribution to investment (Bouakez et al., 2025). Results in Panel B highlights that prior to the mid-2000s, firms receiving either type of spending operated in industries with a similar average investment contribution of approximately 1.5%. Since then, however, industries where firms receive services-related spending have shown a higher contribution to investment compared to those where firms receive goods-related spending.

Panel C informs us about interaction between the two spending categories and the role in the production network. We measure the importance of each sector in the production network by constructing the Katz–Bonacich measure of centrality.⁴⁰ More central sectors are located upstream in the network and supply most of the intermediate inputs to the other industries. By contrast, sectors with low measures of centrality are intermediate-input users and are located downstream. Procurement contracts awarded to firms that operate in downstream industries typically generate higher output responses because the effects propagate upstream through demand for inputs (Baqee and Farhi, 2019; Flynn et al., 2021). Our empirical analysis suggests that firms that receive service procurement contracts tend to operate in industries that are more upstream. Furthermore, this gap has become more marked over time.

Panel D shows the labor intensity of the industries receiving the two types of spending. Labor intensity may cause variations in local fiscal multipliers stemming from goods and services spending, as the shock’s transmission to household budgets can vary (Nakamura and Steinsson, 2014; Alonso, 2017). In labor-intensive sectors, government spending largely benefits workers. A surge in public purchases boosts product demand, prompting firms to increase hiring. Consequently, labor and personal incomes grow, fueling higher consumer spending. Hall (2009) and Bouakez et al. (2025) propose different mechanisms that generate larger output responses when the recipient sector is more labor-intensive. In a setting with a one-sector economy and decreasing returns to scale, Hall (2009) emphasizes that aggregate spending shocks directed to more labor-intensive sectors generate higher multipliers because labor intensity mitigates the substitution effect on labor. Bouakez et al. (2025) show in a multi-sector economy that more labor-intensive sectors have higher

⁴⁰We follow Acemoglu et al. (2012) in calculating the Katz–Bonacich measure of centrality as $\frac{\alpha_H}{S} (\mathbf{I} - \alpha_H \mathbf{W}')^{-1} \mathbf{1}$, with α_H being the average gross-output intensity of intermediate inputs, S the number of sectors, and $\mathbf{W} = \{\omega_{H,sx}\}_{s,x=1}^S$ the BEA Input–Output matrix.

multipliers because they are less distorted, as they rely more on labor, which is the non-marked-up input. By contrast, in standard neoclassical models with inelastic capital supply in the short run, fiscal shocks directed to sectors with lower capital shares (i.e., more labor-intensive) lead to greater price increases and more crowding out, generating smaller multipliers. We measure industry labor intensities by assessing the employee compensation’s contribution to value-added.⁴¹ However, there’s a potential issue due to imperfect competition. Industries in which firms underpay their employees due to monopsony power might be assigned a lower level of labor intensity. To shed light on the role of imperfect competition, we regress labor intensity against employment concentration ratios, used as proxy for imperfect competition.⁴² We find a weak negative correlation between employment concentration and industry labor intensity, suggesting no significant correlation between monopsony power and the measure of labor intensity. Results in Panel D emphasize that the average labor intensity of industries in which firms receiving services spending contracts operate is higher than the average intensity of industries receiving goods spending contracts. Notably, until the mid-2000s, the labor intensity of the industries receiving the two types of spending was similar around 0.65. Starting from the mid-2000s, the labor intensity of industries receiving services contracts has remained constant, while the labor intensity of industries receiving goods contracts has dropped drastically to 0.5.

Panel E explores the degree of competition in the product markets. Imperfect competition in the product market, measured by markups, can influence the effectiveness of fiscal policy. On the one hand, in sectors with high markups, which indicate lower competition and more concentration, the fiscal multiplier is larger because firms with more market power can adjust their prices and output levels more easily in response to fiscal shocks (Rotemberg and Woodford, 1999) or because high-markup firms have a greater influence on aggregate demand and supply (Baqee and Farhi, 2020). On the other hand, Bouakez et al. (2025) argue that the response of aggregate output is larger when spending originates in sectors with lower distortions because the labor wedge relative to the efficient allocation is smaller. We calculate markups using the strategy proposed by Hall (2018), which utilizes sector-level data on output and production inputs from the BEA. The empirical results from Panel E illustrates that sectors receiving goods spending have higher markups than those receiving services spending. The gap between these two is substantial, with the industries receiving goods spending having markups around 1.4, and the industries receiving services spending having markups around 1.1. Furthermore, the gap between the two sets of industries has increased over time. These findings imply that goods are manufactured in less competitive industries.

The last panel of Figure 7 reports the degree of price rigidity. When a fiscal shock occurs in a sector characterized by price stickiness (or low frequency of price adjustments), the interplay between fiscal and monetary policy can enhance the effect of this shock (Cox et al., 2024). Specifically, when government expenditure is directed towards sectors with sticky prices, there is a reduced need for monetary policy to tighten to maintain stable inflation. This results decreases crowding out, thereby enhancing the fiscal

⁴¹The BEA collects detailed Annual data on value-added and employee compensation spanning 66 NAICS codes from 1997 onward.

⁴²The Census Bureau provides employment concentration ratios by industry for 2002 and 2017. In the regressions, we control for year and industry effects.

multiplier effect. Typically, the measure of price stickiness is derived from BLS-restricted microdata by calculating the average frequency of price changes in sectors. We use the industry data on frequency price adjustments calculated by [Pasten et al. \(2020\)](#).⁴³ The empirical patterns show that goods spending is awarded to contractors in sectors where prices change more frequently. Indeed, contractors receiving goods spending operate in industries with a monthly frequency of price changes of 15%. By contrast, services spending contracts are in sectors with a monthly frequency of price adjustments of approximately 10%. Under the Calvo pricing assumption, these results imply that firms receiving goods spending change prices on average every 6.2 months, and firms receiving services spending every 9.5 months.⁴⁴

The results from Figure 7 highlight marked differences in characteristics between the sectors that receive goods spending and the sectors that receive services spending. Firms receiving services spending are in sectors characterized by higher contribution to final consumption and investment, more centrality in the production network, higher labor intensity, lower markups, and lower frequency of price adjustment than the sectors in which firms receiving goods spending operate.

As the producers of goods and services operate in industries with different characteristics, these differences may affect the transmission of a fiscal shock, generating different multiplier effects for the two types of spending. To investigate the relevance of these industry characteristics, we follow a strategy similar to [Bouakez et al. \(2025\)](#). We test the importance of the industry characteristics in explaining the differences in the fiscal multiplier between goods and services spending by running the following specification:

$$\begin{aligned} \frac{v_{l,t+k} - v_{l,t-1}}{v_{l,t-1}} = & \beta_g^k \frac{G_{l,t+k}^g - G_{l,t-1}^g}{Y_{l,t-1}} + \beta_s^k \frac{G_{l,t+k}^s - G_{l,t-1}^s}{Y_{l,t-1}} \\ & + \gamma^k \frac{G_{l,t+k} - G_{l,t-1}}{Y_{l,t-1}} \cdot \frac{(\mathcal{C}_{l,t-1} - \bar{\mathcal{C}})}{\hat{\mathcal{C}}} + \psi \mathcal{C}_{l,t-1} + \alpha_l^k + \delta_{t+k} + \varepsilon_{l,t+k} \end{aligned} \quad (5)$$

where v is an outcome of interest in location l at horizon $t + k$. The interaction term is constructed by multiplying the normalized changes in total spending by the demeaned weighted average of one of the sectoral characteristics. $\mathcal{C}_{l,t}$ is computed at the MSA level by summing, across all sectors, the product of each sector's characteristic and the share of spending directed to that sector in MSA l at time t , relative to the total spending received by that MSA in a given year. $\bar{\mathcal{C}}$ and $\hat{\mathcal{C}}$ are the average and the standard deviation of $\mathcal{C}_{l,t}$ over all MSAs and all years, respectively.⁴⁵

On the one hand, the coefficients β_g and β_s are interpreted as the local fiscal multiplier for an MSA whose weighted-average sectoral characteristic of total spending is equal to the average in the sample. In other words, β_g and β_s measure the response in one of the outcome variables in an MSA with sectoral characteristic

⁴³The frequency price adjustments are measured at the input-output industry codes. We use the BEA concordance tables between NAICS codes and input-output industry codes to recover frequency price adjustments at the NAICS codes. A few NAICS codes are not matched.

⁴⁴The average duration is calculated as $-\frac{1}{\log(1-FPA)}$ with FPA being the mean monthly frequency of price adjustments.

⁴⁵Our specification differs from that of [Bouakez et al. \(2025\)](#) in two aspects. First, we decompose the total expenditures into the first two terms to capture the heterogeneous multiplier effect between goods and services purchases. Second, [Bouakez et al. \(2025\)](#) interact the change in total spending with the sectoral characteristic at the horizon $t + k$. Implicitly, they assume that the evolution of sectoral characteristics is exogenous to the spending shock. In our specification, we interact the spending shock with the sectoral characteristic at $t - 1$, making it pre-determined at the time of the shock.

Table 2: Impulse Responses - Sectoral Characteristics at $t + 3$

	Baseline	Contribution to Consumption	Contribution to Investment	Centrality	Labor Intensity	Markups	Frequency of Price Adjustments	All
<i>Panel A: Employment</i>								
Services Spending	0.867*** (0.224)	0.887*** (0.176)	0.846*** (0.157)	0.897*** (0.176)	0.900*** (0.182)	0.842*** (0.200)	0.879*** (0.174)	0.806*** (0.174)
Goods Spending	0.078 (0.136)	-0.091 (0.079)	-0.063 (0.067)	-0.161 (0.122)	-0.086 (0.060)	-0.008 (0.411)	-0.065 (0.070)	0.020 (0.376)
Spending * Contribution to Consumption		0.012 (0.069)						0.012 (0.082)
Spending * Contribution to Investment			0.095 (0.163)					0.066 (0.127)
Spending * Centrality				-0.149 (0.183)				-0.222* (0.135)
Spending * Labor Intensity					-0.035 (0.063)			-0.070 (0.057)
Spending * Markup						-0.042 (0.214)		-0.091 (0.166)
Spending * Frequency of Price Adjustment							-0.042 (0.080)	-0.072 (0.074)
Observations	12,024	5,997	6,002	6,003	6,007	6,005	5,992	5,984
$\beta_s - \beta_g$	0.789	0.978	0.908	1.058	0.986	0.850	0.944	0.786
Test $\beta_s = \beta_g$	[.002]	[.000]	[.000]	[.000]	[.000]	[.136]	[.000]	[.120]
<i>Panel B: Labor Income</i>								
Services Spending	1.528*** (0.272)	2.025*** (0.422)	1.984*** (0.405)	2.080*** (0.424)	2.058*** (0.434)	1.959*** (0.426)	2.020*** (0.433)	1.872*** (0.357)
Goods Spending	0.129 (0.186)	-0.076 (0.175)	-0.056 (0.129)	-0.375 (0.235)	-0.077 (0.116)	0.044 (0.869)	-0.045 (0.134)	0.049 (0.763)
Spending * Contribution to Consumption		0.002 (0.167)						0.075 (0.174)
Spending * Contribution to Investment			0.090 (0.382)					0.100 (0.226)
Spending * Centrality				-0.615 (0.386)				-0.806*** (0.295)
Spending * Labor Intensity					-0.069 (0.116)			-0.108 (0.109)
Spending * Markup						-0.066 (0.452)		-0.256 (0.332)
Spending * Frequency of Price Adjustment							-0.066 (0.135)	-0.100 (0.134)
Observations	12,024	5,997	6,002	6,003	6,007	6,005	5,992	5,984
$\beta_s - \beta_g$	1.399	2.101	2.039	2.455	2.135	1.915	2.065	1.823
Test $\beta_s = \beta_g$	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[.101]	[0.000]	[.066]
<i>Panel C: GDP</i>								
Services Spending	1.820*** (0.544)	1.574*** (0.526)	1.974*** (0.560)	1.947*** (0.456)	1.796*** (0.533)	1.522 (0.945)	1.775*** (0.549)	1.391 (1.008)
Goods Spending	0.171 (0.269)	0.373 (0.360)	0.074 (0.203)	-0.286 (0.287)	0.168 (0.266)	0.552 (1.454)	0.234 (0.287)	0.531 (1.326)
Spending * Contribution to Consumption		-0.410 (0.301)						-0.115 (0.280)
Spending * Contribution to Investment			-0.417 (0.461)					-0.082 (0.397)
Spending * Centrality				-0.843** (0.422)				-1.064** (0.424)
Spending * Labor Intensity					0.033 (0.143)			-0.149 (0.154)
Spending * Markup						-0.200 (0.696)		-0.428 (0.711)
Spending * Frequency of Price Adjustment							-0.122 (0.179)	-0.150 (0.171)
Observations	4,676	4,669	4,672	4,672	4,674	4,673	4,665	4,660
$\beta_s - \beta_g$	1.648	1.201	1.900	2.233	1.628	0.969	1.541	0.860
Test $\beta_s = \beta_g$	[.005]	[.078]	[.001]	[.000]	[.006]	[.676]	[.013]	[.705]

Notes: The panel includes 344 MSAs. Standard errors are clustered by MSA. The symbols *, **, and *** represent the 10%, 5%, and 1% significance levels, respectively. All regressions include MSA and year-fixed effects.

$\mathcal{C}_{l,t-1} = \bar{\mathcal{C}}$ following a dollar increase in goods and services spending in that MSA, respectively. On the other hand, the coefficient of the interaction term informs us on how the local fiscal multiplier varies with each sectoral characteristic. Our specification implies that there are no heterogeneous effects of the sectoral characteristics on the local fiscal multiplier by category. Equivalently, sectoral characteristics affect the coefficients of the two categories of spending only through the degree of exposure to these characteristics.

Table 2 reports the estimates from equation (5) at the horizon $t + 3$, which represents the peak of the multiplier effects.⁴⁶ The first column reports the baseline estimate for the horizon $t + 3$ from Table B.2. The next six columns present estimates obtained when considering each sectoral characteristic individually, while the last column incorporates all six characteristics simultaneously. The top panel shows the estimates for employment, the middle panel for labor income, and the bottom panel for GDP.

Before investigating to what extent the estimates for goods and services spending change when we control for sectoral characteristics, we explore the role of these characteristics in determining the response to the total government procurement spending. We find that for all three outcomes, the responses are stronger when the spending is directed to sectors with lower markups and centrality, and less frequent price adjustments. Results about contribution to consumption and investment and labor intensity are less clear, as the sign of the estimates varies depending on the outcome variable we analyze. To better assess the quality of our estimates, we compare the results in Panel C of Table 2 with the estimates from Bouakez et al. (2025). We find that our estimates are qualitatively consistent with theirs, such as output responses are stronger when the spending is directed to sectors with lower contribution to consumption and investment, lower centrality, higher labor intensity, lower markups, and less frequent price adjustments.⁴⁷ We document some differences in the magnitude and significance of the coefficients for two reasons. First, we exploit variation at a more granular geographic aggregation, MSAs instead of states. Second, our outcome variables are aggregated at the MSA level rather than per capita.

As the results for the three outcome variable are qualitatively similar, we only discuss the results for GDP reported in Panel C of Table 2. The interaction terms become statistically significant only when we interact the growth rate of the total spending with the measure of centrality. Nevertheless, the difference between the estimates for goods and services spending remain statistically significant. By contrast, the estimate for services spending turns statistically non-significant when we include markup (column 6) and all sectoral characteristics (column 8), making the differences between the services and goods estimates statistically non-significant.⁴⁸ The difference between β_s and β_g drops as well as when we control for the contribution to final consumption, and frequency of price adjustments. This result suggests that markup plays a central role in driving the differences between the services and goods multipliers. Specifically, firms engaged in goods

⁴⁶The literature also commonly uses the horizon $t + 2$ as a reference point. We show the results at the horizon $t + 2$ in the Table B.9. The results for the two horizons are qualitatively similar.

⁴⁷As measure of price rigidity, Bouakez et al. (2025) report the Calvo probability, while we report the frequency of price adjustments, which is one minus the Calvo probability. Hence, a negative estimate in our specification implies a positive coefficient in their regression.

⁴⁸One reason the test fails to reject the null hypothesis of equality between the two estimates is that the reduction in sample size decreases the precision of the coefficient estimates.

production tend to operate in industries characterized by higher markups. This market distortion attenuates the fiscal multiplier associated with government procurement, thereby reducing the responsiveness of output to government purchases of goods more than to purchases of services.

To summarize, this subsection highlights that contractors receiving services contracts operate in industries with different sectoral characteristics —such as contributions to consumption and investment, centrality, labor intensity, markups, and price rigidity —compared to those receiving goods contracts. A larger share of service purchases compared to good purchases is allocated to producers in industries with a larger contribution to final consumption and investment, higher centrality, and higher labor intensity. Conversely, a greater share of good purchases is channeled towards contractors operating in sectors with higher markups and a higher frequency of price adjustments. Our empirical findings suggest that markups play a central role in explaining the differences of the local multiplier responses to goods and services purchases. Nevertheless, part of the differences between the estimates for goods and services purchases remains, particularly for employment and labor income. For this reason, in the next subsections, we explore additional mechanisms that could help explain the remaining unexplained effect after accounting for sectoral characteristics.

6.2 Extensive margin and productivity gains

Government spending may affect firms’ entry and exit decisions, and therefore, the initial demand shock can be amplified through the extensive margin. Increases in spending may reduce the uncertainty about future profits and ease credit constraints, leading to higher turnover rates of firms and productivity gains.⁴⁹

Different types of spending may generate different firm dynamics patterns, leading to heterogeneous multiplier effects. Several reasons could explain these different patterns. First, government spending may lower entry costs and ease credit constraints. As goods producers usually face larger entry costs than services producers, spending directed to goods producers may not be sufficient to lower the entry costs. In this scenario, one should observe the mass of firms entering the market after a shock to services spending to be higher than after a shock to goods spending. Second, after each type of spending, new firms may enter sectors that have a different contribution to aggregate productivity.

To dig into the importance of the extensive margin and productivity gains in explaining the different multiplier effects, we proceed as follows. We collect data series on establishment entry and exit from the County Business Patterns at the MSA-sector level. As we cannot construct labor productivity series at the MSA level from 1979 because data are available only from the early 2000s, we follow a different strategy to quantify the effect of establishment entry and exit on productivity. We compute labor productivity at the sector level, dividing sector GDP by its employment using data from the BEA.⁵⁰ We then classify sectors as sectors with high (low) labor productivity growth if their labor productivity growth rates are above (below) the median growth rate across the sectors. Finally, we aggregate the establishment entry and exit to get

⁴⁹Recent empirical studies have provided insights into the relationship between fiscal policy, business dynamism, and growth. Ferraz et al. (2015) show that firms that win government procurement contracts grow more than their competitors. Lewis and Winkler (2017) show that net firm entry rises after an expansion in US government spending. Bisgaard-Larsen et al. (2025) find positive effects on TFP and labor productivity after government spending shocks.

⁵⁰Data with a consistent definition of sectors are available from 1997 onward.

Table 3: Entry and Exit Rates

	(1) t	(2) t+1	(3) t+2	(4) t+3	(5) t+4
<i>Entry Rate - Sectors High Growth</i>					
Services Spending	2.297** (1.082)	2.586 (1.576)	2.612** (1.197)	2.254*** (0.841)	2.212*** (0.540)
Goods Spending	-0.129 (0.392)	-0.334 (0.297)	-0.183 (0.226)	-0.054 (0.168)	-0.035 (0.175)
<i>Entry Rate - Sectors Low Growth</i>					
Services Spending	-0.611 (0.436)	-0.484 (0.363)	-0.109 (0.401)	0.314 (0.446)	0.493 (0.428)
Goods Spending	0.023 (0.110)	0.339** (0.163)	0.263** (0.115)	0.273* (0.145)	0.406** (0.168)
<i>Exit Rate - Sector High Growth</i>					
Services Spending	-0.681 (1.263)	1.753* (1.029)	2.217** (0.975)	2.457** (1.083)	2.241*** (0.704)
Goods Spending	0.565 (0.400)	0.331 (0.241)	0.296** (0.136)	0.089 (0.147)	0.192 (0.164)
<i>Exit Rate - Sector Low Growth</i>					
Services Spending	-0.261 (0.399)	0.079 (0.403)	0.294 (0.330)	0.681* (0.388)	0.804** (0.344)
Goods Spending	0.256 (0.265)	0.227 (0.158)	0.308** (0.124)	0.392** (0.180)	0.335** (0.169)

Notes: The panel consists of 12,024 observations and includes 344 MSAs. Standard errors are clustered by MSA. The symbols *, **, and *** represent the 10%, 5%, and 1% significance levels, respectively. All regressions include MSA and year-fixed effects.

MSA series of entry and exit in sectors with high or low labor productivity growth. This strategy allows us to explore whether entry and exit occur in sectors that have a greater contribution to productivity.⁵¹

Table 3 reports the estimates for establishment entry and exit by sector's labor productivity growth. Both types of spending incentivize business turnover. We document that service purchases boost the establishment's entry and exit into sectors with high labor productivity growth. By contrast, goods purchases increase the entry and exit rates in sectors with low labor productivity growth, consistent with the fact that goods purchases are directed to contractors in higher markup sectors. These results suggest that service purchases generate an aggregate improvement in productivity larger than goods purchases because firm turnover mostly occurs in high-growth sectors. These results provide suggestive evidence that the extensive margin and productivity gains might amplify differently the fiscal responses to the two categories of purchases.

Our findings are consistent with Bisgaard-Larsen et al. (2025), which also document increases in net

⁵¹We recognize that recent studies have shown a large dispersion in firms' productivity even in narrow industries. Our data do not allow us to identify if the entry or exit is high or low productivity within their sector defined at the 2-digit NAICS. The implicit assumption is that the entry and exit are close to the average productivity of the sector to which they belong. We are aware of this important caveat in the interpretation of our results.

entry and productivity improvements as a consequence of government spending. With respect to them, we highlight the different responses by the type of spending, and we provide new insights about the mechanisms of how spending shocks impact business dynamism and productivity.

6.3 Tradeability

Goods are more tradeable than services.⁵² If the production of goods occurs in neighboring locations, the differences in the fiscal multipliers might reflect geographic spillovers rather than the actual nature of the spending. This explanation could be relevant in our context due to the well-documented issues in correctly allocating government spending to localities. Our data only contain records on prime contracts, and they do not reflect the amount of subcontracting for basic and intermediate materials and components. A contract is assigned to its place of performance, defined as the place where the product is assembled or processed. If the intermediate steps in production are done by sub-contractors outside the performance location, we would geographically misallocate part of the spending.⁵³

We undertake two strategies to quantify the importance of the geographic allocation of contracts, and their “spillover” effects. First, we investigate whether the government spending allocated to locality l could be used for production in neighboring locations. We implement the following specification to investigate whether military spending shocks in location l have some positive effect on the neighboring locations:

$$\frac{\tilde{Y}_{l,t+k} - \tilde{Y}_{l,t-1}}{\tilde{Y}_{l,t-1}} = \tilde{\beta}_g^k \frac{G_{l,t+k}^g - G_{l,t-1}^g}{\tilde{Y}_{l,t-1}} + \tilde{\beta}_s^k \frac{G_{l,t+k}^s - G_{l,t-1}^s}{\tilde{Y}_{l,t-1}} + \alpha_l^k + \delta_{t+k} + \varepsilon_{l,t+k}, \quad (6)$$

where $\tilde{Y}_{l,t+k}$ is the outcome variable for the neighboring locations of l .⁵⁴ We define the neighboring locations to l as MSAs whose center is located within a 50 miles radius from the center of l .⁵⁵

Results in Table 4 show that “spillover” effects in neighboring locations are quite limited, and they do not seem to explain the differences between the local multipliers for the two categories of spending. If any, the estimates point out that there are some positive spillovers coming from service spending shocks in locality l on the neighboring localities rather than from goods spending.⁵⁶

Nevertheless, a significant portion of trade activity extends beyond neighboring MSAs. To capture these

⁵²The average service industry is less tradeable than the average manufacturing industry. However, service tradeability has been growing over time.

⁵³In addition, the definition of the place of performance varies slightly across categories of products. The location of the majority of manufacturing contracts reflects the location of the plant where the product is finally assembled or processed. The location of construction contracts corresponds to the location where the construction is performed. The location for contracts involving purchases from wholesale or other distribution firms reflects the location of the contractor’s place of business. Finally, for service contracts, the location is the place where the service is performed, except for transportation and communication services that report the location of the contracting firm.

⁵⁴Similarly, as before, we instrument the main regressors with a shift-share instrument defined as $Z_{l,t+h}^g = s_l^g \frac{G_{t+k}^g - G_{t-1}^g}{\tilde{Y}_{l,t-1}}$ and $Z_{l,t+h}^s = s_l^s \frac{G_{t+k}^s - G_{t-1}^s}{\tilde{Y}_{l,t-1}}$.

⁵⁵While Auerbach et al. (2020) consider one neighboring location within the distance with a size similar to location l , we include all neighboring MSAs within the distance.

⁵⁶It is worth noticing that the national aggregate fiscal multipliers differ from the local fiscal multipliers. As previously shown by Nakamura and Steinsson (2014) and Chodorow-Reich (2019), the national multiplier accounts for spillover effects. It is beyond the scope of this paper to quantify the national multiplier through the lens of a structural model.

Table 4: Neighboring MSAs

	(1) t	(2) t+1	(3) t+2	(4) t+3	(5) t+4
<i>Panel A: Employment</i>					
Services Spending	-0.028 (0.048)	0.054 (0.112)	0.150 (0.156)	0.194 (0.227)	0.141 (0.224)
Goods Spending	0.020 (0.018)	0.038 (0.060)	0.019 (0.057)	0.029 (0.083)	0.048 (0.095)
<i>Panel B: Labor Income</i>					
Services Spending	-0.062 (0.074)	0.169 (0.193)	0.376 (0.274)	0.534 (0.352)	0.436 (0.341)
Goods Spending	0.031 (0.027)	0.054 (0.083)	0.045 (0.082)	0.056 (0.119)	0.070 (0.128)
<i>Panel C: GDP</i>					
Services Spending	-0.091 (0.063)	0.144 (0.160)	0.516* (0.292)	0.840 (0.512)	0.619 (0.394)
Goods Spending	0.047 (0.034)	0.049 (0.090)	0.036 (0.080)	0.047 (0.135)	0.024 (0.095)

Notes: A neighboring MSA is defined as a MSA within a radius of 50 miles from the comparison MSA. The symbols *, **, and *** represent the 10%, 5%, and 1% significance levels, respectively. All regressions include MSA and year-fixed effects.

interactions, we compare the estimates from the regressions using MSA-level data with those using state-level data. The literature has argued extensively that the geographic misallocation of contracts becomes less important as one moves to larger geographic aggregations. Specifically, it becomes a minor issue at the state level. [Isard \(1962\)](#) argues that geographic disaggregation at the state level does not contain significant measurement errors. [Nakamura and Steinsson \(2014\)](#) use data on shipments from defense industries to the government to verify that, on average, there is a one-for-one relationship between prime contracts allocated to a state and the actual shipments from that state.

The state-level estimates are in Table 5. First, similar to [Demyanyk et al. \(2019\)](#) and [Auerbach et al. \(2020\)](#), the effects are larger when we use the state aggregation. The discrepancy between MSA- and state-level estimates can be attributed to within-state subcontracting. In smaller geographic areas, it is more likely that part of the spending spills into/from other areas. The potential measurement error in subcontracting outside a MSA attenuates the estimates of the local fiscal multipliers, implying the MSA-level estimates to be lower bounds. Second, we do not find any statistically significant effect of goods spending on the outcome variables, while the effects of services spending remain sizable and significant. Nevertheless, we cannot reject the null hypothesis that the estimates for goods and services spending are equal.⁵⁷

These results provide mixed evidence about the role of geographic spillovers in determining the differences

⁵⁷Due to a smaller sample size, the coefficients in Table 5 are estimated less precisely than those in Table B.2, implying that the statistical test may fail to reject the null hypothesis only because of limited sample size.

Table 5: Impulse Responses - State Aggregation

	(1) t	(2) t+1	(3) t+2	(4) t+3	(5) t+4
Panel A: Employment					
Services Spending	0.311 (0.242)	0.696** (0.302)	0.902** (0.389)	1.060** (0.419)	1.042** (0.494)
Goods Spending	0.086 (0.166)	0.059 (0.282)	0.040 (0.359)	0.291 (0.516)	0.760 (0.577)
$\beta_s^k - \beta_g^k$	0.225	0.637	0.862	0.769	0.282
Test $\beta_s^k = \beta_g^k$	[.542]	[.208]	[.196]	[.346]	[.759]
Panel B: Labor Income					
Services Spending	0.384 (0.334)	0.992** (0.488)	1.216* (0.613)	1.443* (0.806)	1.308 (0.946)
Goods Spending	0.163 (0.262)	0.338 (0.502)	0.539 (0.628)	1.068 (0.927)	1.604 (0.968)
$\beta_s^k - \beta_g^k$	0.221	0.654	0.677	0.375	-0.297
Test $\beta_s^k = \beta_g^k$	[.685]	[.440]	[.529]	[.800]	[.855]
Panel C: GDP					
Services Spending	-0.214 (0.540)	0.768* (0.448)	1.223* (0.631)	1.518* (0.806)	1.348 (0.973)
Goods Spending	-0.069 (0.421)	-0.131 (0.698)	0.318 (0.831)	0.871 (1.130)	1.481 (1.172)
$\beta_s^k - \beta_g^k$	-0.145	0.899	0.904	0.646	-0.133
Test $\beta_s^k = \beta_g^k$	[.863]	[.343]	[.474]	[.698]	[.943]

Notes: The panel consists of 1,800 observations and includes 50 states for the period 1979 – 2019. Standard errors are clustered by state. The symbols *, **, and *** represent the 10%, 5%, and 1% significance levels, respectively. All regressions include state and year-fixed effects.

in the estimates for goods and services spending. We cannot completely rule out that geographic spillovers contribute to explaining the gap between the estimates for the two categories of spending.

6.4 Location-specific characteristics

Figure 2 emphasizes that the allocation of spending in the two categories is uneven across locations, with some receiving a relatively larger share of spending in one of the two categories than the other. The differential geographical dispersion of goods-producing and services-producing firms may present a challenge for isolating the differential effects of goods-versus-services spending from differential effects arising from economic differences in the geographies in which spending occurs. Indeed, if locality-specific characteristics are correlated with the probability of receiving a type of spending and the determination of the size of the fiscal multiplier, then the different effects between services and goods spending that we have documented may be caused by some location-specific characteristics.

In the main specification in equation (2), we include location fixed-effects that capture the time-invariant characteristics of a specific location. Thus, our estimates isolate the effects of the different types of spending after absorbing location-specific confounders that may affect the outcome variable of interest.⁵⁸

⁵⁸Furthermore, in constructing the sample, we apply a series of commonly-used filters, described in Section 2, that remove

To further dig into this potential explanation, we test the correlation between the allocation of spending types and the location-specific characteristics. Juarros (2020) documents that DOD spending multipliers are higher with larger shares of small firms.⁵⁹ If locations that receive more services spending also have a larger share of small firms, the larger multiplier effect coming from services spending could be attributed to the firm-size distribution rather than the type of spending. Similarly, if locations receiving more spending on services are more populated, the larger local market may generate larger multiplier effects. To rule out these hypotheses, we compute the correlation between the year-location share of spending on services and location characteristic. The resulting correlation is virtually zero. Overall, location-specific characteristics seem not to be relevant confounders of the multiplier differences between services and goods spending.

7 Conclusions

The Great Recession renewed interest in the effectiveness of fiscal policy. Most of the studies that estimate local fiscal multipliers find positive effects of fiscal spending on outcomes like employment, income, and output. However, there is substantial heterogeneity in the estimates. Theory suggests that local economic characteristics and the composition of government purchases matter for the size of the fiscal multiplier. While a growing empirical literature shows that local economic characteristics can amplify or attenuate the fiscal multiplier, much less is known about the role of the composition of government spending. This paper aims to understand this potential avenue for heterogeneity in fiscal multipliers.

We explore the differences in multipliers generated by two types of government purchases: goods and services. The breakdown into goods versus services is a natural distinction because the US economy has seen two significant structural transformations related to the role of services: the increasing share of services in the consumer basket and the increasing use of services as intermediate inputs.

We show that purchases of services generate positive and significant multiplier effects, while shocks to goods spending have a smaller and non-significant impact. We find that differences in markups, business turnover, and tradeability are important drivers of these multiplier differences.

Our findings suggest that there is room for governments to redesign their fiscal spending and obtain higher multipliers by reallocating dollars from goods toward services. In this respect, our paper reinforces the idea that fiscal authorities should design policy interventions by not only choosing how much to spend but also what to purchase.

outliers locations from our sample and make the locations more comparable along their characteristics.

⁵⁹We collect data on the share of small firms from the Business Dynamics Statistics files. Small firms are defined as firms with less than 20 employees.

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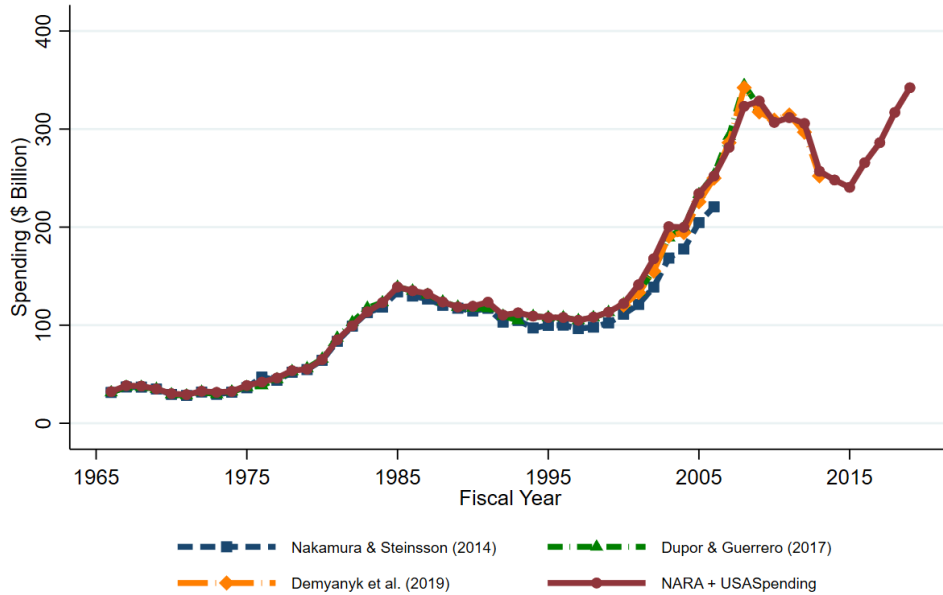
Appendix

A. Data

A.1. Military Spending Validation

This section discusses the quality of the DoD military procurement contract microdata, and we compare them with data previously used in the literature. Figure A.1 compares aggregate military spending in nominal terms at the national level by fiscal years. We compare aggregate military procurement spending derived from our microdata with those calculated by Nakamura and Steinsson (2014), Dupor and Guerrero (2017), and Demyanyk et al. (2019). Overall, we conclude our data closely match the trend and the level of total spending used in previous works.

Figure A.1: Military Spending Comparisons



Note: The period for the comparison matches the research design of the comparison studies. Data for the comparison studies have been downloaded from their journal's data repository.

We also compare the spending in different geographically-disaggregated areas. To this end, we regress

$$Spend_{l,t}^{ours} = \beta Spend_{l,t}^{comp} + \alpha_l + \delta_t + \varepsilon_{l,t}$$

where $Spend_{l,t}^{ours}$ represents the military spending from our data in locality l at time t ; $Spend_{l,t}^{comp}$ is the spending for one of the comparison datasets; and α_l and δ_t are locality and time fixed effects, respectively.

Table A.1 reports the results of our comparison. In column 1, we disaggregate spending at the state level and compare our calculations with the data used by Nakamura and Steinsson (2014). Column 2 shows the

comparison between our state-level aggregation and the data constructed by [Dupor and Guerrero \(2017\)](#). Finally, in column 3, we show the CBSA-level comparison between our data and [Demyanyk et al. \(2019\)](#).

Table A.1: Military Spending Comparisons by Geography

	(1)	(2)	(3)
	Nakamura and Steinsson (2014)	Dupor and Guerrero (2017)	Demyanyk et al. (2019)
β	1.12	0.94	0.98
95% C. I. for β	(1.02 - 1.21)	(0.89 - 0.99)	(0.89 - 1.07)
Observations	2,050	2,200	10,636
Geographic Unit	State	State	CBSA
Number Localities	50	50	862
Period	1966 – 2006	1966 – 2009	2000 – 2012
Locality FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
<i>Within</i> – R^2	0.97	0.96	0.79

Notes: The values in the brackets report the 95% confidence interval. Standard errors are clustered at the geographic unit level. The period of analysis and the geographic aggregation are chosen to match the research design of the comparison studies. Data for the comparison studies have been downloaded from their journal's data repository.

We focus on two tests to evaluate the quality of the geographic distribution of our data compared to previous studies. First, if there were a one-to-one relationship between the geographic allocation, between our data and previously used data, β should be equal or close to one. Second, if there were a strong similarity between our data and the others, the within- R^2 should be high. The results show that the value of one is either included in the 95% confidence interval or it is close to either the upper or lower bound of our estimates of β . The largest discrepancy is between our data and that from [Nakamura and Steinsson \(2014\)](#). This discrepancy, as showed in Figure A.1, comes from the years between 2000 and 2006. The data collected by [Nakamura and Steinsson \(2014\)](#) underestimate total spending, while ours are similar to the other sources of comparison. The within- R^2 is over 0.95 for the comparison with [Nakamura and Steinsson \(2014\)](#) and [Dupor and Guerrero \(2017\)](#), and it is a little bit lower in the comparison with [Demyanyk et al. \(2019\)](#). Because our data are constructed using NARA until 2006, and USASpending and [Demyanyk et al. \(2019\)](#) use only USASpending starting from 2001, these tests also confirm the comparability between the information provided by NARA and USASpending. Overall, these results suggest our data are highly comparable with that used in previous studies.

A.2. Product Codes Classification

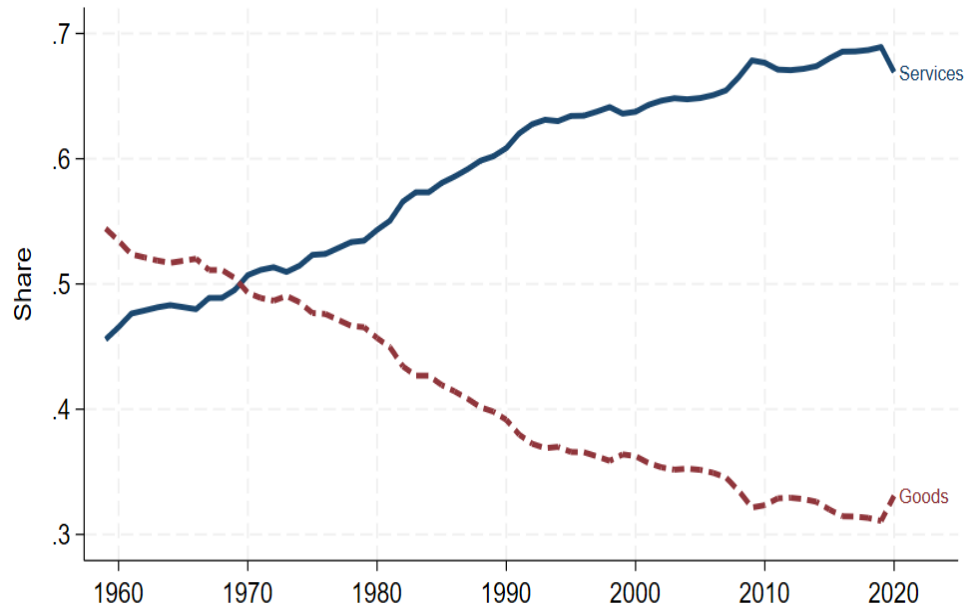
Our dataset includes 3239 distinct 4-digit product codes, of which 2547 are Product Services Codes (PSCs), identifying spending in services, and 692 are the Federal Supply Codes (FSCs), identifying spending in goods. PSCs are grouped into 24 sub-groups, while FSCs are grouped into 85 groups. Table A.2 reports the list of macro-groups for the two categories of spending.

Table A.2: List of Product Codes by Category of Spending

Product Service Codes (PSCs)	Federal Supply Codes (FSCs)
A - Research and Development	10 - Weapons
B - Special Studies/analysis, Not R&D	11 - Nuclear Ordnance
C - Architect/engineer Services	12 - Fire Control Equipment
D - IT and Telecommunication - Information Technology and Telecommunications	13 - Ammunition and Explosives
E - Purchase Of Structures/facilities	14 - Guided Missiles
F - Natural Resources Management	15 - Aerospace Craft and Structural Components
G - Social Services	16 - Aerospace Craft Components and Accessories
H - Quality Control, Test, Inspection	17 - Aerospace Craft Launching, Landing, Ground Handling and Servicing Equipment
J - Maintenance, Repair, Rebuild Equipment	19 - Ships, Small Craft, Pontoon, Docks
K - Modification Of Equipment	20 - Ship and Marine Equipment
L - Technical Representative Services.	22 - Railway Equipment
M - Operation Of Govt Owned Facility	23 - Motor Vehicles, Cycles, Trailers
N - Installation Of Equipment	24 - Tractors
P - Salvage Services	25 - Vehicular Equipment Components
Q - Medical Services	26 - Tires and Tubes
R - Support Services (Prof. Admin, Management)	28 - Engines and Turbines and Component
S - Utilities and Housekeeping	29 - Engine Accessories
T - Photo, Map, Print, Publication	30 - Mechanical Power Transmission Equipment
U - Education and Training	31 - Bearings
V - Transport, Travel, Relocation	32 - Woodworking Machinery and Equipment
W - Lease/rent Equipment	34 - Metalworking Machinery
X - Lease/rent Facilities	35 - Service and Trade
Y - Construct Of Structures/facilities	36 - Special Industry Machinery
Z - Maintenance, Repair, Alter Real Property	37 - Agricultural Machinery and Equipment
	38 - Construct/mine/excavate/highway Equipment
	39 - Materials Handling Equipment
	40 - Rope, Cable, Chain, Fittings
	41 - Refrigeration, Air Condition/circulation Equipment
	42 - Fire/rescue/safety; Environment Protect
	43 - Pumps and Compressors
	44 - Furnace/steam/drying; Nuclear Reactor
	45 - Plumbing, Heating, Waste Disposal
	46 - Water Purification/sewage Treatment
	47 - Pipe, Tubing, Hose, and Fittings
	48 - Valves
	49 - Maintenance/repair Shop Equipment
	51 - Hand Tools
	52 - Measuring Tools
	53 - Hardware and Abrasives
	54 - Prefab Structures/scaffolding
	55 - Lumber, Millwork, Plywood, Veneer
	56 - Construction and Building Material
	58 - Comm/detect/coherent Radiation
	59 - Electrical/electronic Equipment Components
	61 - Electric Wire, Power Distribution Equipment
	62 - Lighting Fixtures, Lamps
	63 - Alarm, Signal, Security Detection
	65 - Medical/dental/veterinary Equipment/supply
	66 - Instruments and Laboratory Equipment
	67 - Photographic Equipment
	68 - Chemicals and Chemical Products
	69 - Training Aids and Devices
	7A - IT and Telecommunication - Applications
	7B - IT and Telecommunication - Compute
	7C - IT and Telecommunication - Data Center
	7D - IT and Telecommunication - Delivery
	7E - IT and Telecommunication - End User
	7F - IT and Telecommunication - IT Management
	7G - IT and Telecommunication - Network
	7H - IT and Telecommunication - Platform
	7J - IT and Telecommunication - Security and Compliance
	7K - IT and Telecommunication - Storage
	71 - Furniture
	72 - Household/commercial Furnish/appliance
	73 - Food Preparation/serving Equipment
	74 - Office Mach/text Process/visib Rec
	75 - Office Supplies and Devices
	76 - Books, Maps, Other Publications
	77 - Musical Inst/phonograph/home Radio
	78 - Recreational/athletic Equipment
	79 - Cleaning Equipment and Supplies
	80 - Brushes, Paints, Sealers, Adhesives
	81 - Containers/packaging/packing Suppl
	83 - Textile/leather/fur; Tent; Flag
	84 - Clothing, Individual Equipment, Insigna, and Jewelry
	85 - Toiletries
	87 - Agricultural Supplies
	88 - Live Animals
	89 - Subsistence
	91 - Fuels, Lubricants, Oils, Waxes
	93 - Nonmetallic Fabricated Materials
	94 - Nonmetallic Crude Materials
	95 - Metal Bars, Sheets, Shapes
	96 - Ores, Minerals and Primary Products
	99 - Miscellaneous

B. Additional Empirical Results

Figure B.1: Private Consumption Expenditures by Category



Note: Data on the private consumption expenditures by goods and services have been downloaded from the Bureau of Economic Analysis.

Table B.1: Impulse Responses - Total Spending

	(1) t	(2) t+1	(3) t+2	(4) t+3	(5) t+4
<i>Panel A: Employment</i>					
Spending	0.092* (0.050)	0.245** (0.114)	0.268** (0.126)	0.361* (0.192)	0.384** (0.181)
F-stat	9.684	22.946	15.961	11.037	12.369
<i>Panel B: Labor Income</i>					
Spending	0.134** (0.064)	0.467*** (0.163)	0.505*** (0.188)	0.657** (0.289)	0.652** (0.270)
F-stat	9.684	22.946	15.961	11.037	12.369
<i>Panel C: GDP</i>					
Spending	0.055 (0.038)	0.452*** (0.148)	0.404** (0.164)	0.559* (0.329)	0.462** (0.220)
F-stat	11.921	37.565	17.718	5.578	10.854

Notes: The panel includes 344 MSAs and consists of 12,024 observations for the period 1979 – 2019 for Panels A and B, and 4,676 observations for the period 2001 – 2019 for Panel C. Standard errors are clustered by MSA. The symbols *, **, and *** represent the 10%, 5%, and 1% significance levels, respectively. All regressions include MSA and year-fixed effects.

Table B.2: Impulse Responses - Spending by Components

	(1) t	(2) t+1	(3) t+2	(4) t+3	(5) t+4
Panel A: Employment					
Services Spending	0.246** (0.124)	0.560*** (0.191)	0.722*** (0.212)	0.867*** (0.224)	0.771*** (0.179)
Goods Spending	0.033 (0.033)	0.075 (0.101)	0.065 (0.101)	0.078 (0.136)	0.118 (0.142)
S-W F-stat Services	14.435	86.299	65.444	42.787	40.807
S-W F-stat Goods	21.199	41.414	32.187	30.537	53.276
$\beta_s^k - \beta_g^k$	0.214	0.486	0.657	0.789	0.654
Test $\beta_g^k = \beta_s^k$	[.111]	[.033]	[.008]	[.002]	[.004]
Panel B: Labor Income					
Services Spending	0.307*** (0.081)	0.960*** (0.166)	1.315*** (0.200)	1.528*** (0.272)	1.315*** (0.248)
Goods Spending	0.047 (0.048)	0.157 (0.139)	0.126 (0.139)	0.129 (0.186)	0.143 (0.183)
S-W F-stat Services	14.435	86.299	65.444	42.787	40.807
S-W F-stat Goods	21.199	41.414	32.187	30.537	53.276
$\beta_s^k - \beta_g^k$	0.260	0.804	1.189	1.399	1.173
Test $\beta_g^k = \beta_s^k$	[.011]	[.001]	[.000]	[.000]	[.000]
Panel C: GDP					
Services Spending	0.303** (0.135)	1.031*** (0.270)	1.406*** (0.309)	1.820*** (0.544)	1.592*** (0.438)
Goods Spending	-0.007 (0.048)	0.275 (0.225)	0.118 (0.173)	0.171 (0.269)	0.103 (0.187)
S-W F-stat Services	6.582	47.196	31.130	4.128	7.758
S-W F-stat Goods	18.892	88.286	93.962	23.699	42.772
$\beta_s^k - \beta_g^k$	0.310	0.756	1.288	1.648	1.489
Test $\beta_g^k = \beta_s^k$	[.045]	[.040]	[.000]	[.005]	[.002]

Notes: The panel includes 344 MSAs and consists of 12,024 observations for the period 1979 – 2019 for Panels A and B, and 4,676 observations for the period 2001 – 2019 for Panel C. Standard errors are clustered by MSA. The symbols *, **, and *** represent the 10%, 5%, and 1% significance levels, respectively. All regressions include MSA and year-fixed effects.

Table B.3: Cumulative Local Fiscal Multiplier - Spending by Components

	(1) t	(2) t+1	(3) t+2	(4) t+3	(5) t+4
<i>Panel A: Employment</i>					
Services Spending	0.246** (0.124)	0.485*** (0.173)	0.637*** (0.196)	0.780*** (0.205)	0.806*** (0.190)
Goods Spending	0.033 (0.033)	0.065 (0.084)	0.075 (0.096)	0.091 (0.118)	0.115 (0.129)
$\beta_s^k - \beta_g^k$	0.214	0.420	0.563	0.688	0.691
Test $\beta_g^k = \beta_s^k$	[.111]	[.039]	[.014]	[.004]	[.003]
<i>Panel B: Labor Income</i>					
Services Spending	0.307*** (0.081)	0.799*** (0.145)	1.133*** (0.178)	1.392*** (0.209)	1.412*** (0.224)
Goods Spending	0.047 (0.048)	0.132 (0.115)	0.146 (0.133)	0.164 (0.165)	0.179 (0.177)
$\beta_s^k - \beta_g^k$	0.260	0.667	0.987	1.227	1.233
Test $\beta_g^k = \beta_s^k$	[.011]	[.001]	[.000]	[.000]	[.000]
<i>Panel C: GDP</i>					
Services Spending	0.303** (0.135)	0.752*** (0.238)	1.048*** (0.268)	1.342*** (0.317)	1.422*** (0.346)
Goods Spending	-0.007 (0.048)	0.159 (0.143)	0.136 (0.162)	0.155 (0.217)	0.130 (0.207)
$\beta_s^k - \beta_g^k$	0.310	0.593	0.912	1.186	1.291
Test $\beta_g^k = \beta_s^k$	[.045]	[.044]	[.005]	[.002]	[.002]

Notes: The panel includes 344 MSAs and consists of 12,024 observations for the period 1979 – 2019 for Panels A and B, and 4,676 observations for the period 2001 – 2019 for Panel C. Standard errors are clustered by MSA. The symbols *, **, and *** represent the 10%, 5%, and 1% significance levels, respectively. All regressions include MSA and year-fixed effects.

Table B.4: Newly-created Jobs

	(1) t	(2) t+1	(3) t+2	(4) t+3	(5) t+4
<i>Panel A: All Industries</i>					
Services Spending	0.582 (0.294)	1.325 (0.451)	1.708 (0.501)	2.050 (0.529)	1.823 (0.424)
Goods Spending	0.077 (0.077)	0.178 (0.239)	0.154 (0.239)	0.184 (0.322)	0.278 (0.334)
<i>Panel B: Transportation & Utilities</i>					
Services Spending	-0.002 (0.019)	0.025 (0.037)	0.060 (0.041)	0.080 (0.042)	0.095 (0.044)
Goods Spending	-0.003 (0.009)	-0.008 (0.012)	-0.009 (0.008)	-0.012 (0.011)	-0.013 (0.015)
<i>Panel C: Construction</i>					
Services Spending	0.032 (0.028)	0.068 (0.039)	0.095 (0.049)	0.120 (0.058)	0.114 (0.056)
Goods Spending	0.016 (0.013)	0.014 (0.021)	0.012 (0.021)	0.021 (0.032)	0.039 (0.036)
<i>Panel D: Manufacturing</i>					
Services Spending	0.164 (0.050)	0.311 (0.085)	0.439 (0.106)	0.604 (0.131)	0.580 (0.134)
Goods Spending	0.003 (0.032)	0.066 (0.047)	0.092 (0.038)	0.132 (0.050)	0.139 (0.048)
<i>Panel E: Wholesale Trade</i>					
Services Spending	-0.005 (0.022)	0.030 (0.028)	0.033 (0.027)	0.050 (0.036)	0.035 (0.030)
Goods Spending	-0.002 (0.008)	0.000 (0.016)	-0.004 (0.010)	-0.010 (0.012)	-0.016 (0.011)
<i>Panel F: Retail Trade</i>					
Services Spending	0.024 (0.029)	0.121 (0.056)	0.189 (0.081)	0.247 (0.086)	0.241 (0.075)
Goods Spending	0.000 (0.017)	0.012 (0.030)	0.014 (0.027)	0.034 (0.035)	0.048 (0.041)
<i>Panel G: Finance</i>					
Services Spending	0.008 (0.018)	0.043 (0.045)	0.073 (0.052)	0.114 (0.058)	0.125 (0.050)
Goods Spending	-0.009 (0.011)	0.003 (0.018)	0.003 (0.013)	0.023 (0.014)	0.041 (0.016)
<i>Panel H: Services</i>					
Services Spending	0.205 (0.080)	0.594 (0.156)	0.768 (0.134)	0.928 (0.180)	0.945 (0.184)
Goods Spending	-0.082 (0.086)	-0.034 (0.115)	-0.066 (0.132)	-0.058 (0.166)	-0.078 (0.226)

Notes: The estimates for each industry are based on 344 MSAs for the period 1979 – 2019. The standard errors are calculated using the delta method.

Table B.5: Impulse Responses - Different Price Adjustments

	(1) t	(2) t+1	(3) t+2	(4) t+3	(5) t+4
<i>Panel A: Employment</i>					
Services Spending	0.258* (0.142)	0.573** (0.227)	0.752*** (0.250)	0.918*** (0.257)	0.820*** (0.204)
Goods Spending	0.030 (0.027)	0.075 (0.091)	0.061 (0.090)	0.065 (0.120)	0.095 (0.122)
$\beta_s^k - \beta_g^k$	0.227	0.498	0.691	0.852	0.725
Test $\beta_g^k = \beta_s^k$	[.127]	[.052]	[.013]	[.003]	[.002]
<i>Panel B: Labor Income</i>					
Services Spending	0.300*** (0.089)	0.986*** (0.192)	1.399*** (0.233)	1.647*** (0.301)	1.414*** (0.276)
Goods Spending	0.042 (0.040)	0.147 (0.126)	0.109 (0.122)	0.100 (0.161)	0.108 (0.157)
$\beta_s^k - \beta_g^k$	0.258	0.839	1.290	1.547	1.307
Test $\beta_g^k = \beta_s^k$	[.013]	[.001]	[.000]	[.000]	[.000]
<i>Panel C: GDP</i>					
Services Spending	0.349** (0.159)	1.254*** (0.323)	1.722*** (0.370)	2.285*** (0.673)	1.987*** (0.533)
Goods Spending	-0.011 (0.043)	0.203 (0.196)	0.080 (0.146)	0.109 (0.204)	0.059 (0.150)
$\beta_s^k - \beta_g^k$	0.361	1.051	1.642	2.176	1.928
Test $\beta_g^k = \beta_s^k$	[.040]	[.008]	[.000]	[.002]	[.001]

Notes: The panel includes 344 MSAs and consists of 12,024 observations for the period 1979 – 2019 for Panels A and B, and 4,676 observations for the period 2001 – 2019 for Panel C. Standard errors are clustered by MSA. The symbols *, **, and *** represent the 10%, 5%, and 1% significance levels, respectively. All regressions include MSA and year-fixed effects.

Table B.6: Impulse Responses - Excluding top 1% contracts by MSA-year

	(1) t	(2) t+1	(3) t+2	(4) t+3	(5) t+4
<i>Panel A: Employment</i>					
Services Spending	0.761*** (0.166)	1.120*** (0.275)	1.393*** (0.413)	1.489*** (0.450)	1.330*** (0.405)
Goods Spending	0.147 (0.178)	0.325 (0.282)	0.562 (0.415)	0.831 (0.546)	1.254* (0.694)
$\beta_s^k - \beta_g^k$	0.614	0.795	0.831	0.658	0.075
Test $\beta_g^k = \beta_s^k$	[.021]	[.069]	[.183]	[.376]	[.930]
<i>Panel B: Labor Income</i>					
Services Spending	0.871*** (0.188)	1.685*** (0.318)	2.285*** (0.397)	2.395*** (0.434)	2.027*** (0.373)
Goods Spending	0.280 (0.268)	0.518 (0.434)	0.895 (0.681)	1.250 (0.881)	1.762 (1.083)
$\beta_s^k - \beta_g^k$	0.591	1.167	1.390	1.145	0.265
Test $\beta_g^k = \beta_s^k$	[.106]	[.053]	[.100]	[.265]	[.825]
<i>Panel C: GDP</i>					
Services Spending	1.611*** (0.428)	2.904*** (0.823)	3.084*** (1.110)	3.332*** (1.250)	3.189*** (1.227)
Goods Spending	-0.414 (0.687)	-0.426 (0.705)	-0.183 (0.540)	-0.313 (0.635)	-0.472 (0.660)
$\beta_s^k - \beta_g^k$	2.025	3.331	3.267	3.644	3.661
Test $\beta_g^k = \beta_s^k$	[.023]	[.005]	[.015]	[.015]	[.014]

Notes: The panel includes 344. Standard errors are clustered by MSA. The symbols *, **, and *** represent the 10%, 5%, and 1% significance levels, respectively. All regressions include MSA and year-fixed effects.

Table B.7: Impulse Responses - Excluding MSAs with large contracts

	(1) t	(2) t+1	(3) t+2	(4) t+3	(5) t+4
<i>Panel A: Employment</i>					
Services Spending	0.247* (0.132)	0.592*** (0.201)	0.767*** (0.221)	0.931*** (0.229)	0.838*** (0.175)
Goods Spending	0.018 (0.034)	0.039 (0.102)	0.031 (0.103)	0.034 (0.131)	0.061 (0.136)
$\beta_s^k - \beta_g^k$	0.230	0.553	0.736	0.897	0.777
Test $\beta_g^k = \beta_s^k$	[.105]	[.019]	[.004]	[.001]	[.000]
<i>Panel B: Labor Income</i>					
Services Spending	0.299*** (0.082)	0.959*** (0.166)	1.327*** (0.201)	1.567*** (0.295)	1.365*** (0.277)
Goods Spending	0.014 (0.052)	0.084 (0.140)	0.059 (0.139)	0.042 (0.171)	0.040 (0.168)
$\beta_s^k - \beta_g^k$	0.286	0.874	1.267	1.524	1.325
Test $\beta_g^k = \beta_s^k$	[.007]	[.000]	[.000]	[.000]	[.000]
<i>Panel C: GDP</i>					
Services Spending	0.348** (0.152)	1.075*** (0.293)	1.399*** (0.323)	1.862*** (0.618)	1.618*** (0.482)
Goods Spending	0.000 (0.045)	0.282 (0.230)	0.131 (0.176)	0.197 (0.285)	0.125 (0.195)
$\beta_s^k - \beta_g^k$	0.348	0.793	1.268	1.665	1.493
Test $\beta_g^k = \beta_s^k$	[.038]	[.036]	[.001]	[.010]	[.004]

Notes: The panel includes 199 MSAs. Standard errors are clustered by MSA. The symbols *, **, and *** represent the 10%, 5%, and 1% significance levels, respectively. All regressions include MSA and year-fixed effects.

Table B.8: Impulse Responses - Controlling for the Average Contract Size

	(1) t	(2) t+1	(3) t+2	(4) t+3	(5) t+4
<i>Panel A: Employment</i>					
Services Spending	0.241** (0.121)	0.546*** (0.185)	0.704*** (0.206)	0.848*** (0.220)	0.759*** (0.178)
Goods Spending	0.034 (0.031)	0.083 (0.095)	0.074 (0.098)	0.096 (0.137)	0.129 (0.143)
$\beta_s^k - \beta_g^k$	0.207	0.463	0.630	0.752	0.630
Test $\beta_s^k = \beta_g^k$	[.109]	[.031]	[.008]	[.003]	[.005]
<i>Panel B: Labor Income</i>					
Services Spending	0.302*** (0.078)	0.942*** (0.157)	1.284*** (0.190)	1.497*** (0.261)	1.296*** (0.241)
Goods Spending	0.048 (0.046)	0.168 (0.127)	0.142 (0.131)	0.159 (0.182)	0.159 (0.184)
$\beta_s^k - \beta_g^k$	.254	.774	1.143	1.338	1.137
Test $\beta_s^k = \beta_g^k$	[.009]	[.000]	[.000]	[.000]	[.000]
<i>Panel C: GDP</i>					
Services Spending	0.306** (0.136)	1.048*** (0.272)	1.411*** (0.309)	1.813*** (0.540)	1.604*** (0.443)
Goods Spending	-0.008 (0.051)	0.229 (0.233)	0.094 (0.181)	0.134 (0.255)	0.074 (0.194)
$\beta_s^k - \beta_g^k$	0.314	0.820	1.317	1.679	1.531
Test $\beta_s^k = \beta_g^k$	[.046]	[.029]	[.000]	[.004]	[.002]

Notes: The panel includes 344 MSAs. Standard errors are clustered by MSA. The symbols *, **, and *** represent the 10%, 5%, and 1% significance levels, respectively. All regressions include MSA and year-fixed effects.

Table B.9: Impulse Responses - Sectoral Characteristics at $t + 2$

	Baseline	Contribution to Consumption	Contribution to Investment	Centrality	Labor Intensity	Markups	Frequency of Price Adjustments	All
<i>Panel A: Employment</i>								
Services Spending	0.722*** (0.212)	0.758*** (0.111)	0.830*** (0.117)	0.856*** (0.110)	0.833*** (0.123)	0.810*** (0.233)	0.773*** (0.113)	0.988*** (0.344)
Goods Spending	0.065 (0.101)	-0.019 (0.044)	-0.048 (0.054)	-0.067 (0.060)	-0.030 (0.048)	-0.032 (0.163)	-0.006 (0.048)	-0.142 (0.278)
Spending * Contribution to Consumption		-0.053 (0.043)						-0.166 (0.167)
Spending * Contribution to Investment			-0.085 (0.108)					-0.082 (0.159)
Spending * Centrality				-0.097 (0.086)				0.056 (0.193)
Spending * Labor Intensity					-0.031 (0.064)			-0.211* (0.116)
Spending * Markup						0.005 (0.105)		0.099 (0.226)
Spending * Frequency of Price Adjustment							-0.048 (0.082)	-0.136 (0.084)
Observations	12,024	5,997	6,002	6,003	6,007	6,005	5,992	5,984
$\beta_s - \beta_g$	0.657	0.777	0.878	0.923	0.863	0.842	0.779	1.130
Test $\beta_s = \beta_g$	[.008]	[.000]	[.000]	[.000]	[.000]	[.029]	[.000]	[.064]
<i>Panel B: Labor Income</i>								
Services Spending	1.315*** (0.200)	1.673*** (0.249)	1.821*** (0.259)	1.947*** (0.222)	1.776*** (0.253)	1.775*** (0.477)	1.690*** (0.254)	2.134*** (0.622)
Goods Spending	0.126 (0.139)	0.015 (0.067)	-0.068 (0.096)	-0.165 (0.142)	-0.003 (0.086)	-0.034 (0.350)	0.029 (0.079)	-0.196 (0.493)
Spending * Contribution to Consumption		-0.066 (0.076)						0.045 (0.238)
Spending * Contribution to Investment			-0.260 (0.196)					-0.395 (0.295)
Spending * Centrality				-0.386** (0.155)				-0.335 (0.307)
Spending * Labor Intensity					-0.050 (0.103)			-0.147 (0.180)
Spending * Markup						0.029 (0.218)		-0.021 (0.382)
Spending * Frequency of Price Adjustment							-0.059 (0.128)	-0.160 (0.146)
Observations	12,024	5,997	6,002	6,003	6,007	6,005	5,992	5,984
$\beta_s - \beta_g$	1.189	1.658	1.888	2.111	1.779	1.810	1.661	2.330
Test $\beta_s = \beta_g$	[.000]	[.000]	[.000]	[.000]	[.000]	[.024]	[.000]	[.034]
<i>Panel C: GDP</i>								
Services Spending	1.406*** (0.309)	1.226*** (0.323)	1.593*** (0.332)	1.676*** (0.245)	1.249*** (0.358)	1.491*** (0.531)	1.278*** (0.331)	1.690** (0.808)
Goods Spending	0.118 (0.173)	0.174 (0.121)	-0.021 (0.118)	-0.354 (0.234)	0.205 (0.178)	0.020 (0.615)	0.209 (0.167)	-0.255 (1.024)
Spending * Contribution to Consumption		-0.304* (0.155)						0.045 (0.552)
Spending * Contribution to Investment			-0.499* (0.292)					-0.283 (0.406)
Spending * Centrality				-0.798*** (0.287)				-0.800 (0.502)
Spending * Labor Intensity					0.225 (0.276)			-0.249 (0.463)
Spending * Markup						0.060 (0.309)		-0.080 (0.754)
Spending * Frequency of Price Adjustment							-0.224 (0.236)	-0.361 (0.266)
Observations	4,676	4,669	4,672	4,672	4,674	4,673	4,665	4,660
$\beta_s - \beta_g$	1.288	1.052	1.614	2.030	1.044	1.471	1.068	1.945
Test $\beta_s = \beta_g$	[.000]	[.003]	[.000]	[.000]	[.020]	[.177]	[.008]	[.282]

Notes: The panel includes 344 MSAs. Standard errors are clustered by MSA. The symbols *, **, and *** represent the 10%, 5%, and 1% significance levels, respectively. All regressions include MSA and year-fixed effects.